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Gut Dysbiosis:

Manifestations – Mediators – Mitigators

NICOLE MEIER, ND

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Disclosures

Nicole Meier, ND, disclosed no relevant financial relationships with any ineligible companies.

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Learning Objectives

At the conclusion of this segment, successful participants will be able to:

1. Explain the normal functions of the gut microbiome and the related benefits to the host.

2. List common clinical findings, antecedents, triggers, and mediators of gut dysbiosis.

3. Explain how gut dysbiosis may serve as a mediator of extraintestinal disease and/or dysfunction.

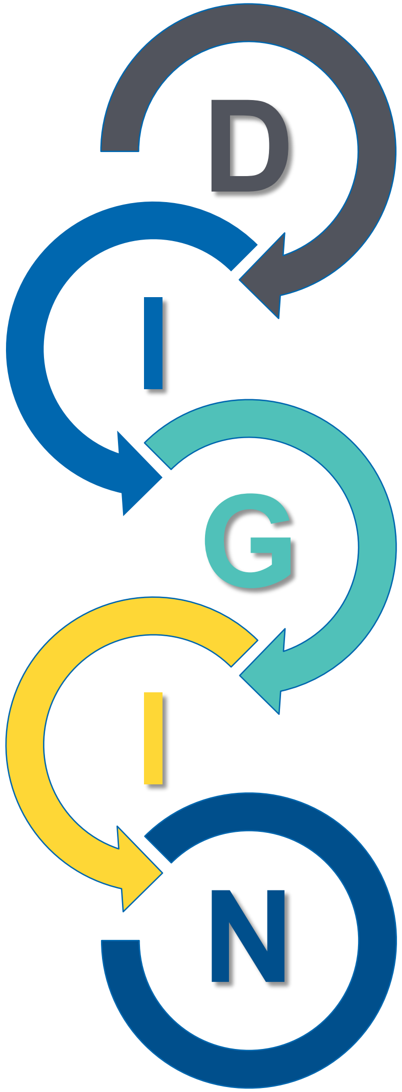
IFM Clinical Skills and Performance Objectives

THIS SEGMENT WILL ENABLE PARTICIPANTS TO DEVELOP & APPLY THESE SKILLS IN THEIR PRACTICE:

1. Identify and document the patient’s clinical findings of gut dysbiosis and elicit the antecedents, triggers, and mediators thereof on IFM's Functional Medicine Timeline and Matrix.

2. Identify extraintestinal diseases and dysfunctions that may be mediated by gut dysbiosis.

The DIGIN Mnemonic of Gut Dysfunctions



***D**igestion/Absorption*

***I**ntestinal Permeability*

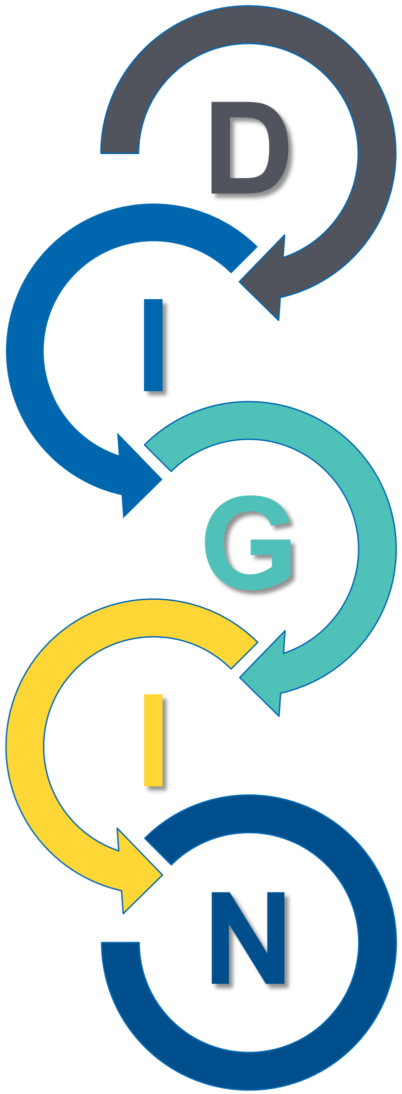
***G**ut Microbiota/Dysbiosis*

***I**nflammation/Immune*

***N**ervous System*

The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS, Patrick Hanaway, MD, and IFM.

The DIGIN Mnemonic of Gut Dysfunctions



***G**ut Microbiota/Dysbiosis*

The DIGIN Model was developed collaboratively by Liz Lipski, PhD, CNS, Patrick Hanaway, MD, and IFM.

Clinical Focus

- Definitions of key terms associated with dysbiosis.

Terminology

- **Microbiota:**
  - The population of micro-organisms in a given location

- **Microbiome:**
  - The genetic component of the microbiota
  - *The environment in which the microbiota lives and influences*

- **Gut Metabolome:**
  - The metabolic activities and metabolites produced by the intestinal microbiota

Liu J, Tan Y, Cheng H, Zhang D, Feng W, Peng C. Functions of gut microbiota metabolites, current status and future perspectives. *Aging Dis*. 2022;13(4):1106-1126. doi:10.14336/AD.2022.0104

Terminology

Beneficial or Symbiotic Species:

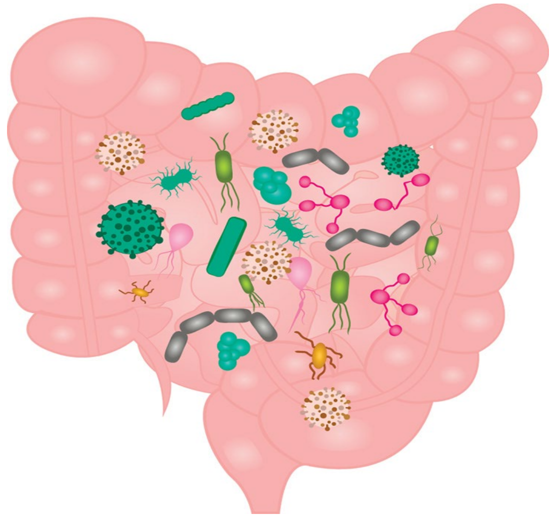
Micro-organisms that provide benefit to the host and benefit from the host

Commensal:

An organism that resides within or upon a host, benefits from the host, and does not harm or benefit the host

Infection (vs Dysbiosis):

An infestation by pathological micro-organisms that causes harm to the host, *the effects of which are mediated by the host's immune system's response to the micro-organism*



Kumamoto CA, Gresnigt MS, Hube B. The gut, the bad and the harmless: *Candida albicans* as a commensal and opportunistic pathogen in the intestine. *Curr Opin Microbiol*. 2020;56:7-15. doi:10.1016/j.mib.2020.05.006

Terminology

Dysbiosis (vs Infection):

- Defined by an imbalance in bacterial composition, changes in bacterial metabolic activities, and/or changes in bacterial distribution within the gut
- Usually refers to colonic dysbiosis
- SIBO = small intestinal bacterial overgrowth

Types of Dysbiosis

- **Deficit:** Reduction or loss of beneficial species
- **Overgrowth:** Excess of potentially harmful microbes
- **Diversity:** Reduced "richness" or abnormal balance of species
- **Distribution:** Presence of microbes in unusual location

Kriss M, Hazleton KZ, Nusbacher NM, Martin CG, Lozupone CA. Low diversity gut microbiota dysbiosis: drivers, functional implications and recovery. *Curr Opin Microbiol*. 2018;44:34-40. doi:10.1016/j.mib.2018.07.003

The Cast of Characters

The Cast of Characters

Types of Microbes in the Human Microbiota

Viruses (acellular): mainly bacteriophages: "virome"

Prokaryotes (without a nucleus)

- Archaea (strict anaerobes)
- Bacteria (aerobes and anaerobes)
  - Bacteroidetes and firmicutes are the dominant phyla

Eukaryotes (nucleus + organelles)

- Microfungi (yeast—Candida, Malassezia, Saccharomyces): "mycobiome"
- Protists (Blastocystis, Dientamoeba, Giardia, Cryptosporidium)
- Helminths (parasitic worms)

- Shreiner AB, Kao JY, Young VB. The gut microbiome in health and in disease. *Curr Opin Gastroenterol*. 2015;31(1):69-75. doi:10.1097/MOG.0000000000000139
- Hou K, Wu ZX, Chen XY, et al. Microbiota in health and diseases. *Signal Transduct Target Ther*. 2022;7(1):135. doi:10.1038/s41392-022-00974-4

Clinical Focus

- Beneficial functions of the gut microbiota and its metabolites.

The Gut Microbiota And Host Health:
A New Clinical Frontier

Like the immune system, the microbiome of the gut is unique in each individual, contains components that are heritable, and contains 150 more genes than the host. Without it, virtually all physiological aspects of the host are altered. The gut microbiome is modifiable through ***diet, antibiotics, stress, chemicals, and other environmental factors***, each **influencing the makeup and diversity of the microbiome and ultimately physiological function**. In all of these ways, the gut microbiome functions like another organ in the body.

Marchesi JR, Adams DH, Fava F, et al. The gut microbiota and host health: a new clinical frontier. *Gut.* 2016;65(2):330-339. doi:10.1136/gutjnl-2015-309990

Microbiota Functions in the Gut

- **Protective Activities:** The barrier effect – resident bacteria provide resistance to colonization by potentially pathogenic microbes
- **Neural Signaling:** Bidirectional interface with insular cortex
- **Metabolic Activities:** Microflora ferments non-digestible dietary residue, releasing SCFAs and numerous vitamins; also provides detoxification capacity

Carabotti M, Scirocco A, Maselli MA, Severi C. The gut-brain axis: interactions between enteric microbiota, central and enteric nervous systems. *Ann Gastroenterol*. 2015;28(2):203-209.

Microbiota Functions in the Gut

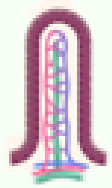
- **Trophic Activities:** SCFAs produced by microflora control epithelial cell proliferation and differentiation in the colon (to protect against the development of neoplasia)

- **Immune Activation:** Education and modulation of the immune system and development of oral tolerance

- Liu G, Tang J, Zhou J, Dong M. Short-chain fatty acids play a positive role in colorectal cancer. *Discov Oncol*. 2024;15(1):425. doi:10.1007/s12672-024-01313-5
- Belkaid Y, Hand TW. Role of the microbiota in immunity and inflammation. *Cell*. 2014;157(1):121-141. doi:10.1016/j.cell.2014.03.011

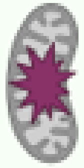
Beneficial Functions of the Gut Microbiome

STRUCTURAL



- ↑ Differentiation & proliferation of epithelial cells
- Intestinal villi & crypts development
- Villus vascularization/angiogenesis
- Tight junction regulation

METABOLISM

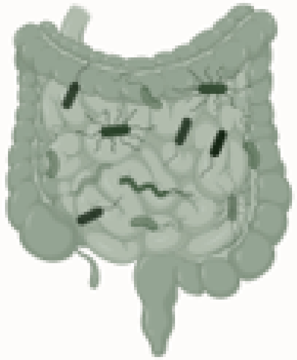


- SCFA production & metabolism
- Glucose metabolism
- ↑ Insulin sensitivity
- ↓ Adipose deposition
- Bone homeostasis
- Bile acid metabolism

BIOSYNTHESIS

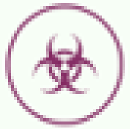
- Hormones (e.g., CCK, 5-HT, ghrelin)
- Neurotransmitters (e.g., serotonin, GABA, catecholamines)
- Neuropeptides, enzymes
- ↑ Antioxidant production

BENEFICIAL GUT MICROBIOME FUNCTIONS



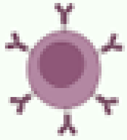
EUBIOSIS

DETOXIFICATION



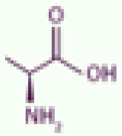
- Metabolism of xenobiotics/toxins
- Drug metabolism - bioavailability/efficacy

IMMUNITY



- GALT maturation & mucosal immune regulation
- Innate & adaptive immune activation
- Immunocompetence/ tolerance
- ↓ Inflammatory mediators

NUTRITION



- Vitamin synthesis: Vit K, thiamine (B1), riboflavin (B2), biotin (B7), folate (B9), pantothenic acid (B5), cobalamin (B12)
- Hydrolysis/fermentation of complex carbohydrates
- Amino acid synthesis

PROTECTION

- ↑ Barrier integrity, mucus layer
- ↑ sIgA
- ↑ Antimicrobial peptides
- ↓ Luminal pH
- ↓ Pathogenic colonization

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See: “References: Gut Microbiome Functions”

Long Description: Beneficial Functions of the Gut Microbiome

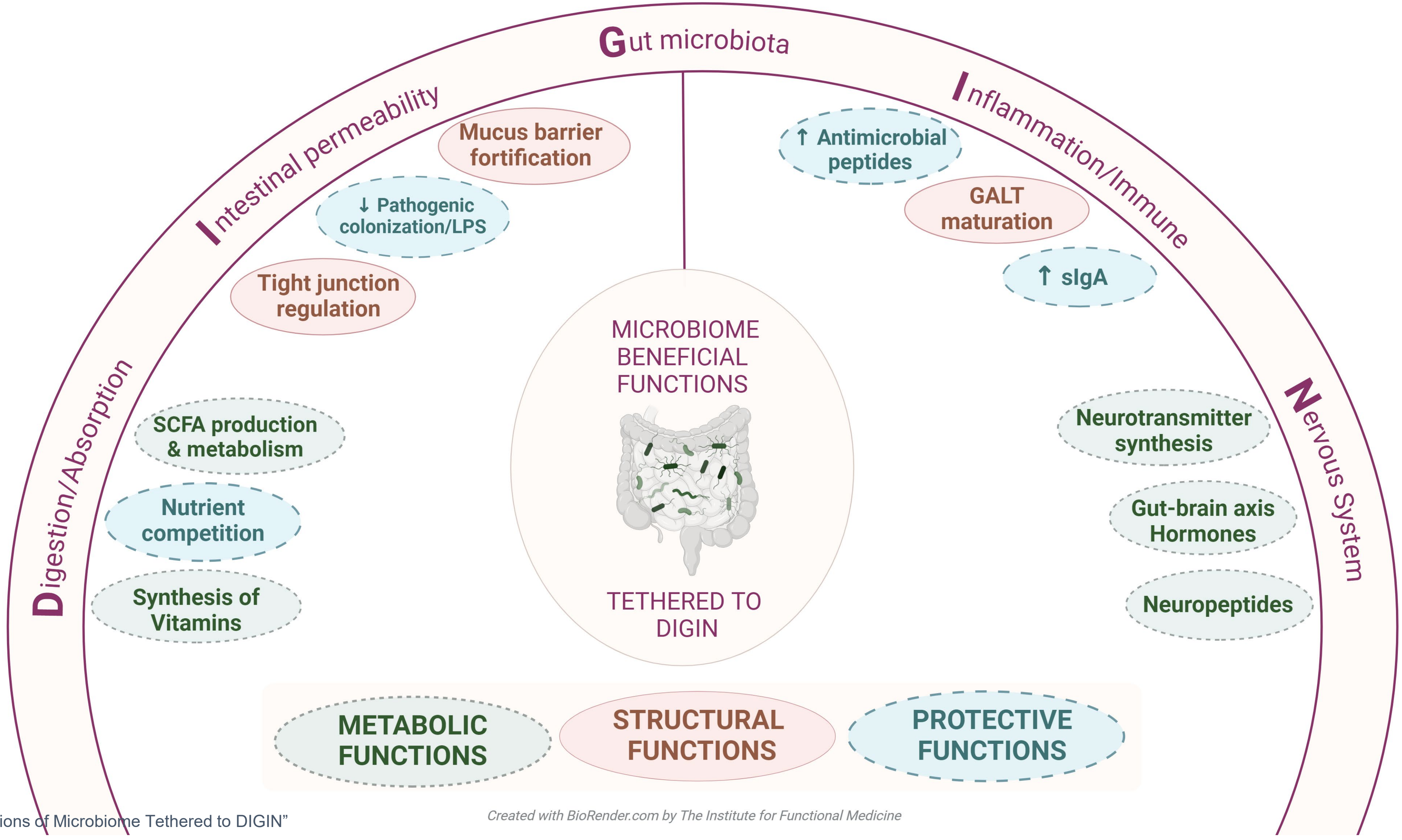
- Damaged villi and damaged cell junctions permit intestinal permeability and malabsorption.

References

GUT MICROBIOME FUNCTIONS

- Afzaal M, Saeed F, Shah YA, et al. Human gut microbiota in health and disease: unveiling the relationship. *Front Microbiol*. 2022;13:999001. doi:10.3389/fmicb.2022.999001
- Adak A, Khan MR. An insight into gut microbiota and its functionalities. *Cell Mol Life Sci*. 2019;76(3):473-493. doi:10.1007/s00018-018-2943-4
- Leeuwendaal NK, Cryan JF, Schellekens H. Gut peptides and the microbiome: focus on ghrelin. *Curr Opin Endocrinol Diabetes Obes*. 2021;28(2):243-252. doi:10.1097/MED.0000000000000616
- Liu R, Hong J, Xu X, et al. Gut microbiome and serum metabolome alterations in obesity and after weight-loss intervention. *Nat Med*. 2017;23(7):859-868. doi:10.1038/nm.4358
- Morowitz MJ, Carlisle EM, Alverdy JC. Contributions of intestinal bacteria to nutrition and metabolism in the critically ill. *Surg Clin North Am*. 2011;91(4):771-viii. doi:10.1016/j.suc.2011.05.001
- Prakash S, Rodes L, Coussa-Charley M, Tomaro-Duchesneau C. Gut microbiota: next frontier in understanding human health and development of biotherapeutics. *Biologics*. 2011;5:71-86. doi:10.2147/BTT.S19099
- Vrieze A, Van Nood E, Holleman F, et al. Transfer of intestinal microbiota from lean donors increases insulin sensitivity in individuals with metabolic syndrome [published correction appears in *Gastroenterology*. 2013;144(1):250]. *Gastroenterology*. 2012;143(4):913-6.e7. doi:10.1053/j.gastro.2012.06.031

Functions of Microbiome Tethered to DIGIN



See: “References: Functions of Microbiome Tethered to DIGIN”

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Long Description: Functions of Microbiome Tethered to D I G I N

Beneficial microbiome functions tethered to D I G I N

- Digestion and absorption
  - Short chain fatty acid production and metabolism
  - Nutrient competition
  - Synthesis of vitamins
- Intestinal permeability
  - Mucus barrier fortification
  - Decreased pathogenic colonization and L P S
  - Tight junction regulation
- Gut Microbiota
- Inflammation and immune
  - Increased antimicrobial peptides
  - GALT maturation
  - Increased secretory I G A
- Nervous system
  - Neurotransmitter synthesis
  - Gut-brain axis hormones
  - Neuropeptides

References

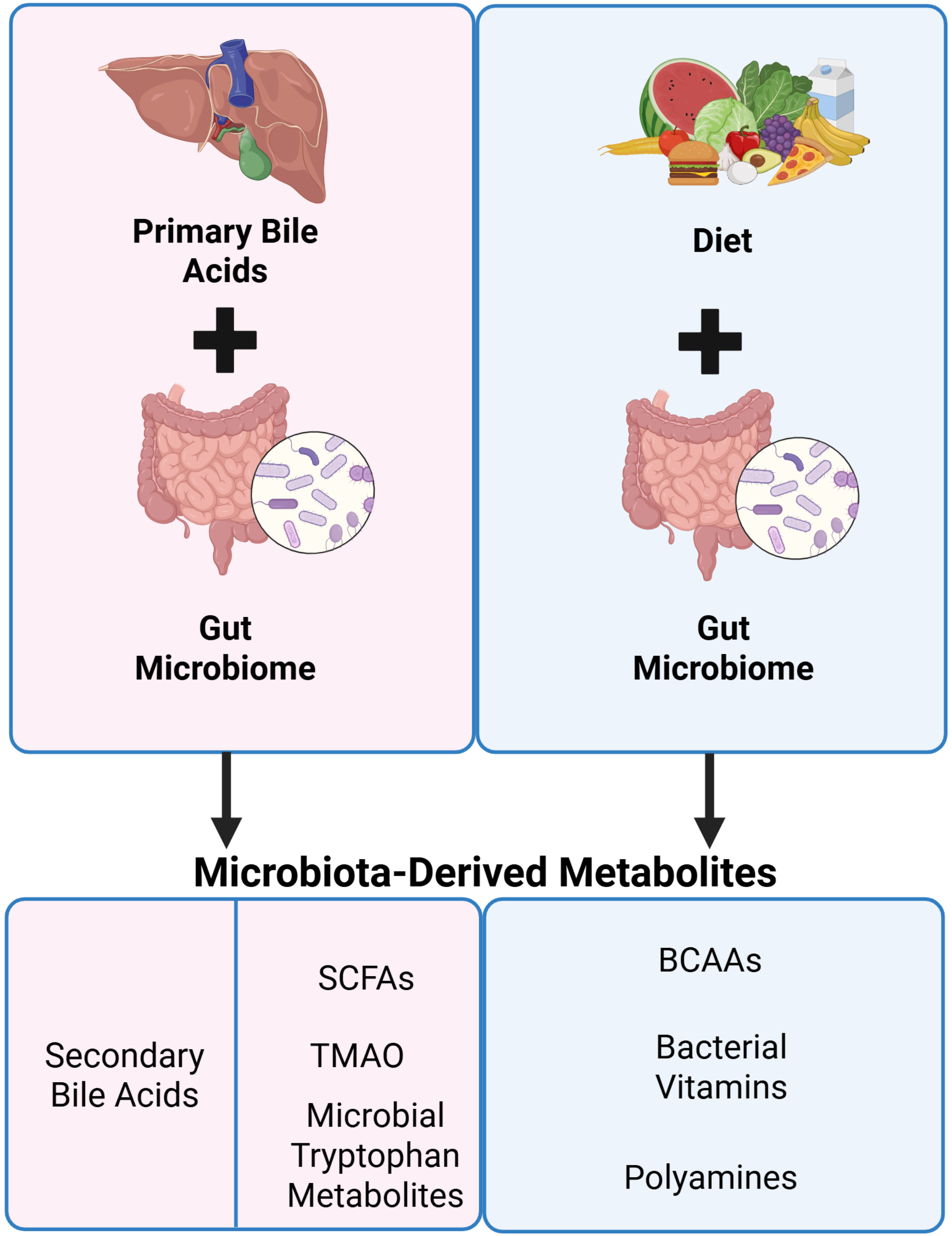
FUNCTIONS OF MICROBIOME TETHERED TO DIGIN

- Adak A, Khan MR. An insight into gut microbiota and its functionalities. *Cell Mol Life Sci*. 2019;76(3):473-493. doi:10.1007/s00018-018-2943-4
- Afzaal M, Saeed F, Shah YA, et al. Human gut microbiota in health and disease: unveiling the relationship. *Front Microbiol*. 2022;13:999001. doi:10.3389/fmicb.2022.999001
- Gomes AC, Hoffmann C, Mota JF. The human gut microbiota: metabolism and perspective in obesity. *Gut Microbes*. 2018;9(4):308-325. doi:10.1080/19490976.2018.1465157
- Wei P, Keller C, Li L. Neuropeptides in gut-brain axis and their influence on host immunity and stress. *Comput Struct Biotechnol J*. 2020;18:843-851. doi:10.1016/j.csbj.2020.02.018

Microbiota-Derived Metabolites

Two Categories of Origin

- Diet
- Secondary Bile Acids



- Created with BioRender.com by The Institute for Functional Medicine
- Liu J, Tan Y, Cheng H, Zhang D, Feng W, Peng C. Functions of gut microbiota metabolites, current status and future perspectives. *Aging Dis*. 2022;13(4):1106-1126. doi:10.14336/AD.2022.0104

Long Description: Microbiota-Derived Metabolites

- Microbiota-derived metabolites can originate from dietary or host-derived substrates that microbiota transform, or they can be produced through de novo synthesis. Microbes can transform primary bile acids into secondary bile acids, fiber into short-chain fatty acids, and protein into branched-chain amino acids. Gut microbiota can synthesize compounds de novo, such as in the case of vitamin K.

Metabolomics: Short-Chain Fatty Acids (SCFAs)

Malabsorbed carbohydrate and nondigestible fibers are fermented by colonic bacteria into short-chain fatty acids.

- Primary fuel for colon
- Absorbed by colonic mucosa & used as energy
- Enhance sodium and water absorption
- Exert trophic effects in small intestine and colon
- Regulate cell proliferation, differentiation, carcinogenesis
- Role in immunomodulation

- den Besten G, van Eunen K, Groen AK, Venema K, Reijngoud DJ, Bakker BM. The role of short-chain fatty acids in the interplay between diet, gut microbiota, and host energy metabolism. *J Lipid Res*. 2013;54(9):2325-2340. doi:10.1194/jlr.R036012
- Fusco W, Lorenzo MB, Cintoni M, et al. Short-chain fatty-acid-producing bacteria: key components of the human gut microbiota. *Nutrients*. 2023;15(9):2211. doi:10.3390/nu15092211

SCFA Production in Colon

Carbohydrates

+

Fiber

Bacterial Enzymes

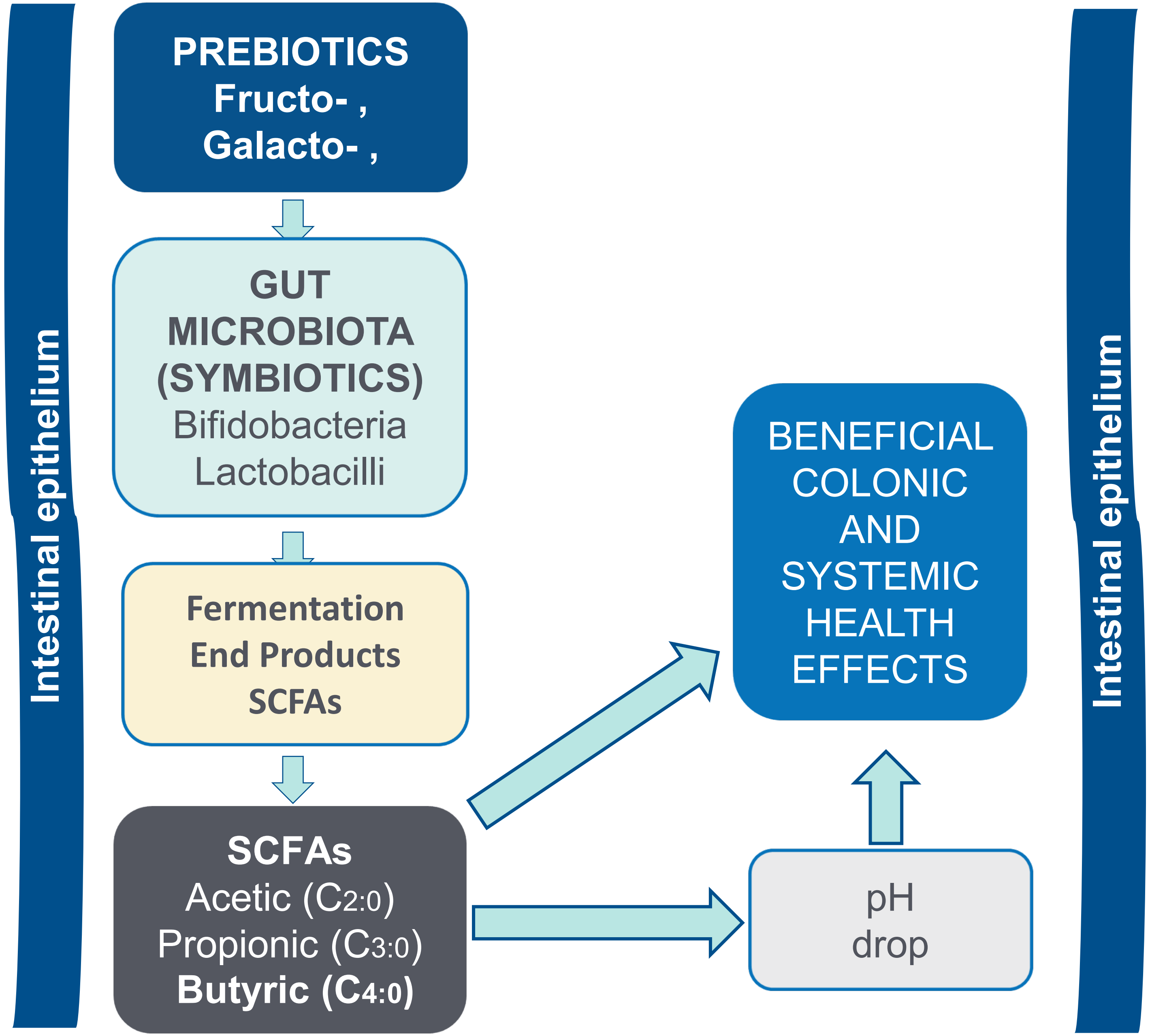
Short-Chain Fatty Acids

- **Butyrate**
- **Acetate**
- **Propionate**

Long Description: S C F A Production in Colon

- Bacterial enzymes act on carbohydrates and fiber to produce short-chain fatty acids, including butyrate, acetate, and propionate.

The Key Role of SCFAs and the Metabolic Functions of the Microbiome



The healthy gut microbiome produces 50-100 mmol·L−1 per day of SCFAs

Adapted from: Huazano-García A, López MG. Metabolism of short chain fatty acids in the colon and faeces of mice after a supplementation of diets with agave fructans. In: Baez RV, ed. *Lipid Metabolism*. IntechOpen; 2013. doi:10.5772/51248. Accessed June 3, 2025. https://www.intechopen.com/books/lipid-metabolism/metabolism-of-short-chain-fatty-acids-in-the-colon-and-faeces-of-mice-after-a-supplementation-of-die. Licensed under CC BY 3.0.

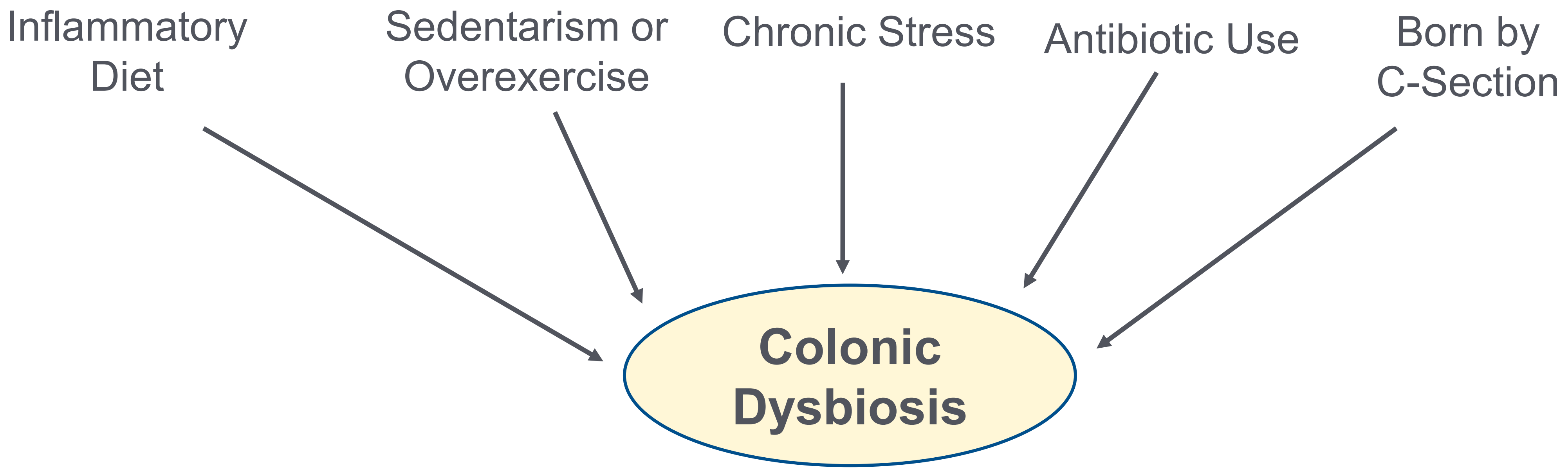
Long Description: The Key Role of S C F As and the Metabolic Functions of the Microbiome

- Prebiotics and gut microbiota produce acetic, propionic, and butyric acids. These short-chain fatty acids lower P H, and exert beneficial colonic and systemic health effects.

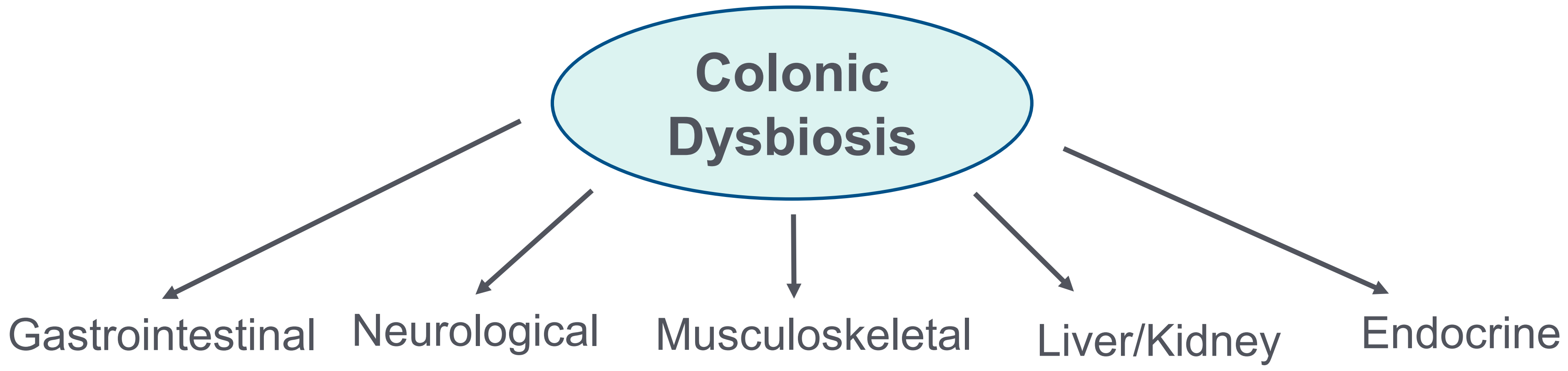
Clinical Focus

- Pathophysiological mechanisms of the effects of gut dysbiosis.

COLONIC DYSBIOSIS – MANY ROOT CAUSES



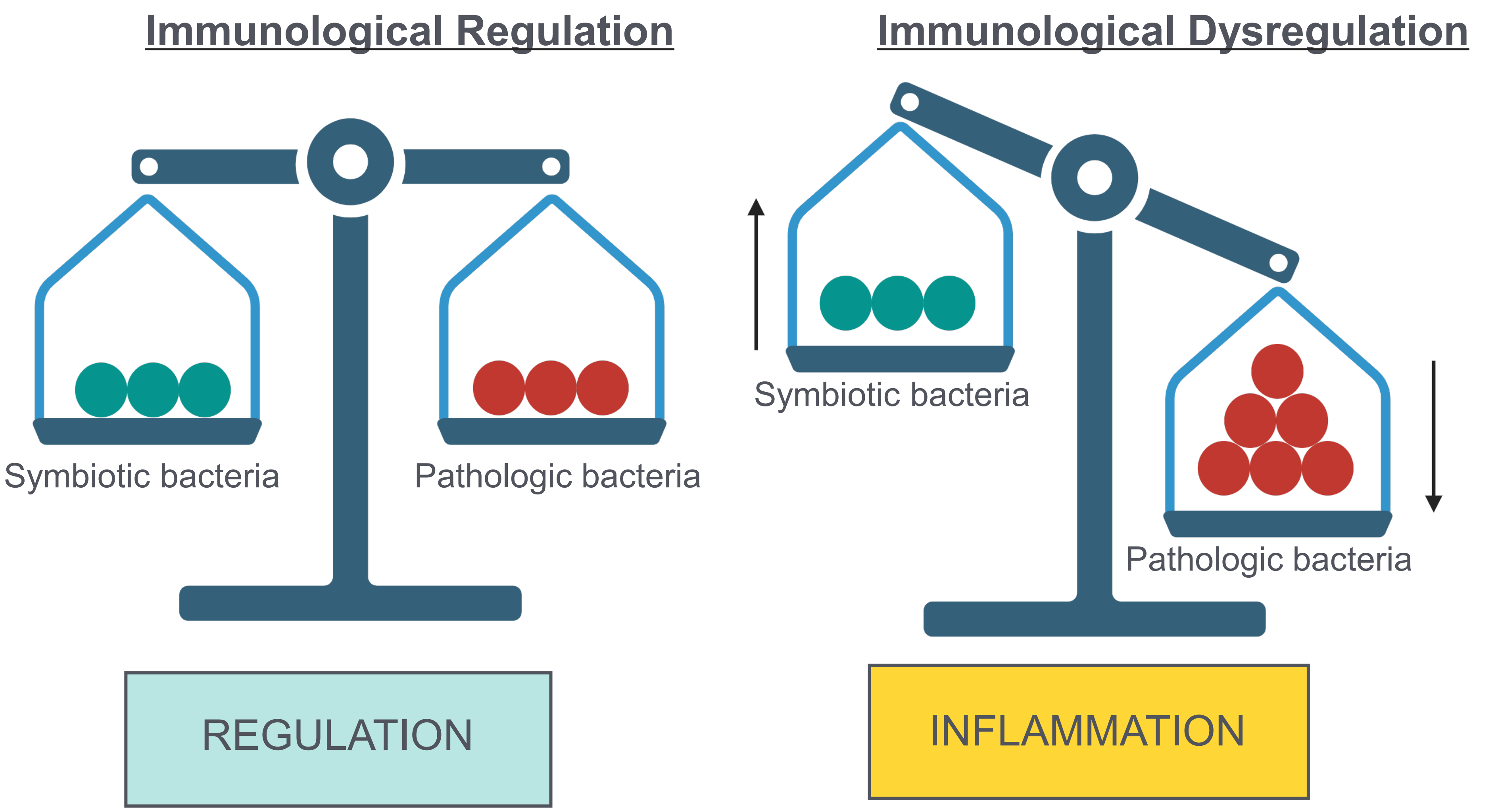
COLONIC DYSBIOSIS – MANY MANIFESTATIONS



Long Description: Colonic Dysbiosis

- Colonic dysbiosis: Many root causes
  - Inflammatory diet
  - Sedentarism or over-exercise
  - Chronic stress
  - Antibiotic use
  - Born by C-section
- Colonic dysbiosis: many manifestations
  - Gastrointestinal
  - Neurological
  - Musculoskeletal
  - Liver, kidney
  - Endocrine

Microbiome Balance

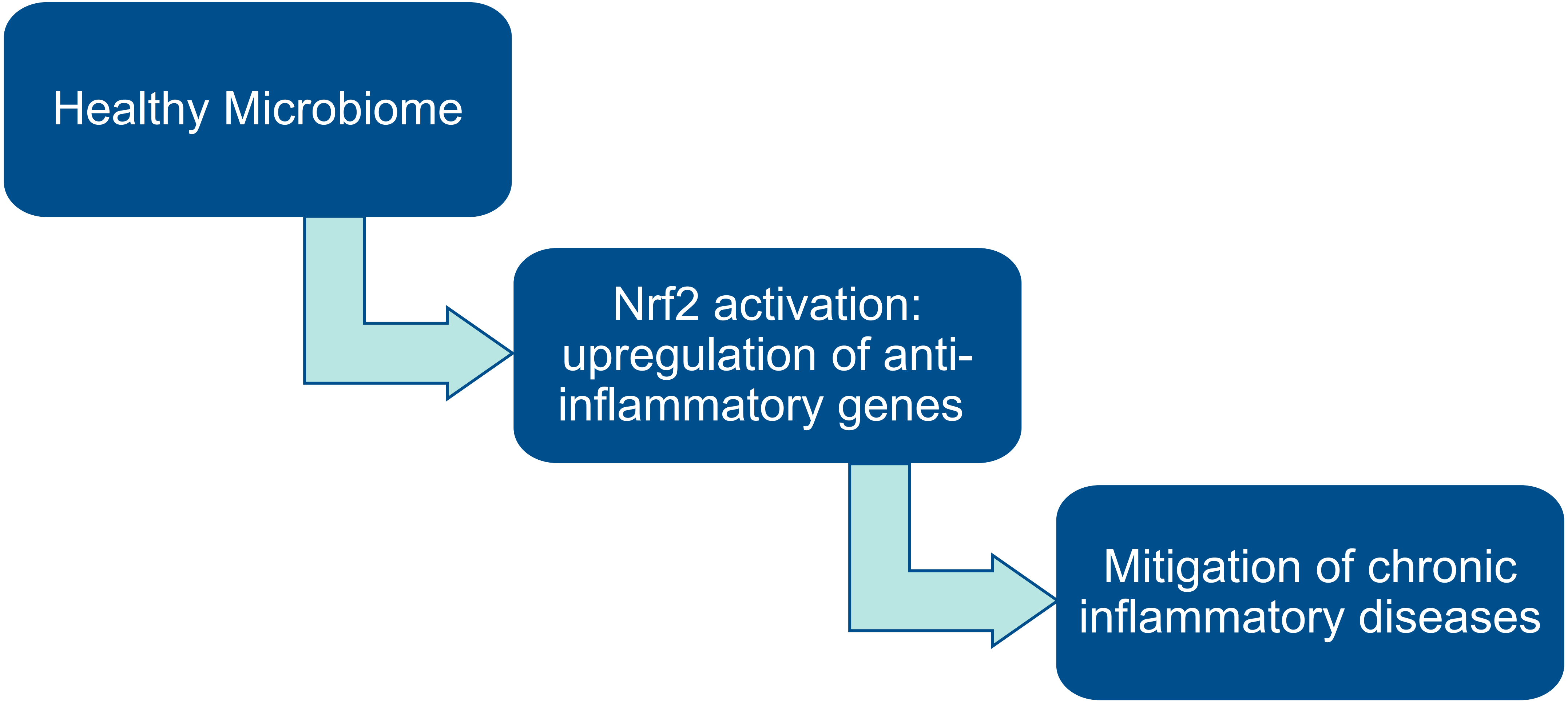


Long Description: Microbiome Balance

- Immunological regulation occurs when there is a balance between symbiotic and pathogenic bacteria, whereas inflammation and immunological dysregulation occur when pathogenic bacteria outweigh symbiotic bacteria.

Gut Microbiome

Favorable Modulation of Gene Transcription/Expression



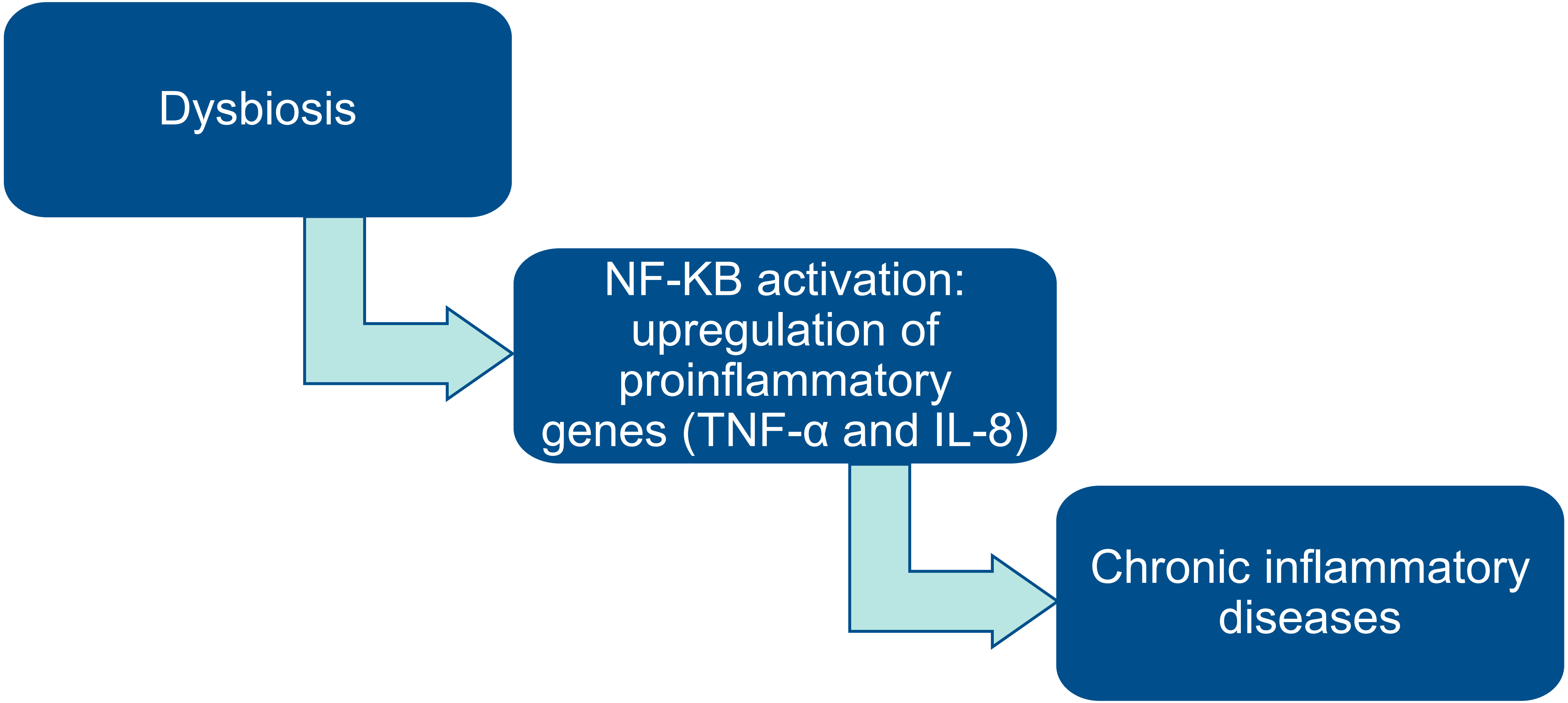
See "References: Gut Microbiome: Modulation of Gene Transcription/Expression"

Long Description: Gut Microbiome Favorable Modulation of Gene Transcription/Expression

- In a healthy microbiome, activation of N R F 2 and the resulting upregulation of antioxidant and cytoprotective genes contribute to the mitigation of chronic inflammatory disease.

Dysbiosis

Unfavorable Modulation of Gene Transcription/Expression



See "References: Dysbiosis: Modulation of Gene Transcription/Expression"

Long Description: Dysbiosis - Unfavorable Modulation of Gene Transcription/Expression

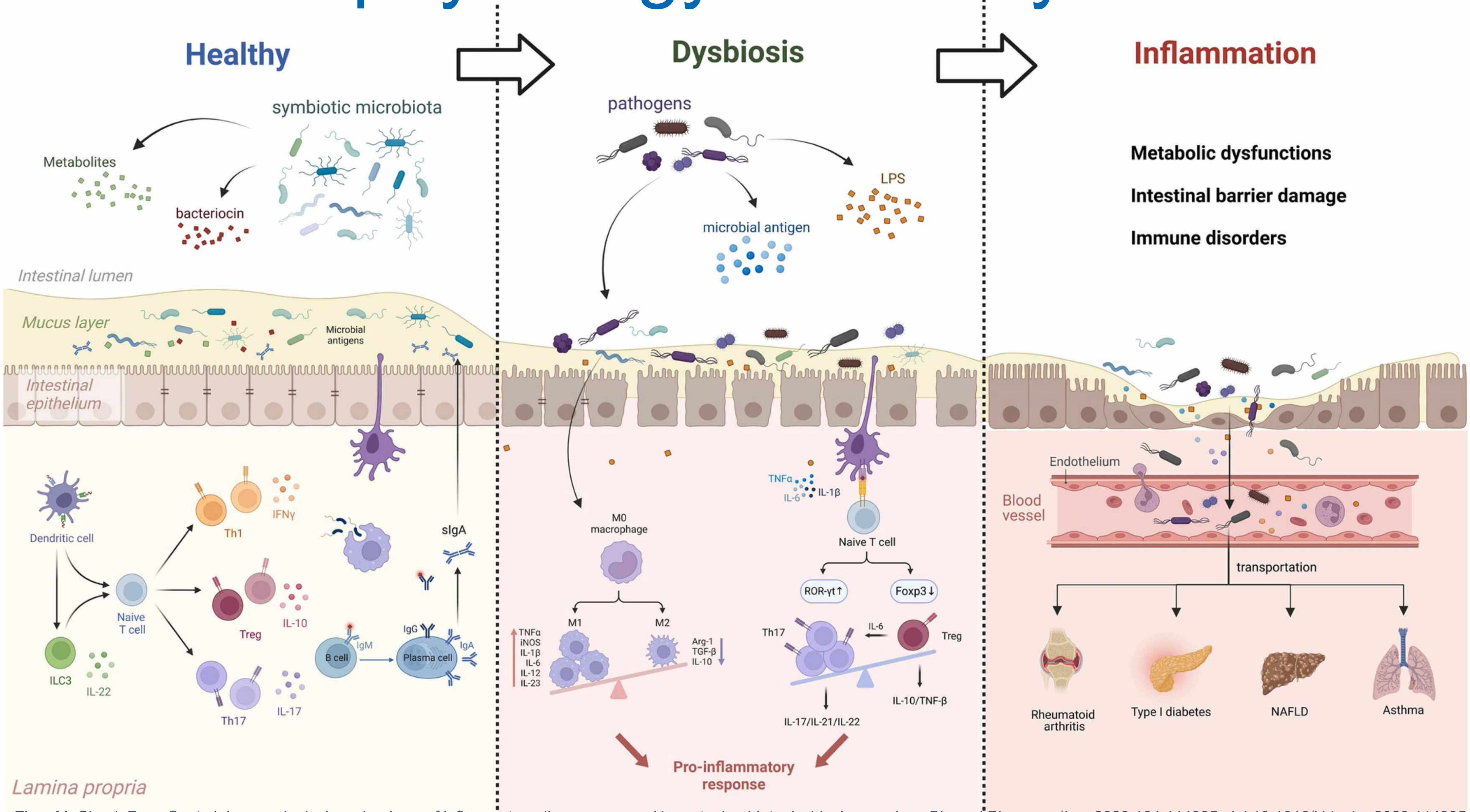
- In dysbiosis, N F kappa B activation and the resulting upregulation of pro-inflammatory genes such as T N F alpha and I L 8 contribute to the promotion of chronic inflammatory disease.

References

DYSBIOSIS: MODULATION OF GENE TRANSCRIPTION/EXPRESSION

- Belizário JE, Faintuch J, Garay-Malpartida M. Gut microbiome dysbiosis and immunometabolism: new frontiers for treatment of metabolic diseases. *Mediators Inflamm*. 2018;2018:2037838. doi:10.1155/2018/2037838
- Mostafavi Abdolmaleky H, Zhou JR. Gut microbiota dysbiosis, oxidative stress, inflammation, and epigenetic alterations in metabolic diseases. *Antioxidants (Basel)*. 2024;13(8):985. doi:10.3390/antiox13080985
- Zhao M, Chu J, Feng S, et al. Immunological mechanisms of inflammatory diseases caused by gut microbiota dysbiosis: a review. *Biomed Pharmacother*. 2023;164:114985. doi:10.1016/j.biopha.2023.114985
- Zheng D, Liwinski T, Elinav E. Interaction between microbiota and immunity in health and disease. *Cell Res*. 2020;30(6):492-506. doi:10.1038/s41422-020-0332-7

Pathophysiology of Gut Dysbiosis



Zhao M, Chu J, Feng S, et al. Immunological mechanisms of inflammatory diseases caused by gut microbiota dysbiosis: a review. *Biomed Pharmacother*. 2023;164:114985. doi:10.1016/j.biopha.2023.114985

Long Description: Pathophysiology of Gut Dysbiosis

- In a healthy intestinal environment, symbiotic microbiota produce bacteriocins and metabolites that promote a healthy immune response and keep pathogenic microbes in balance. In dysbiosis, pathogenic microbes secrete L P S and microbial antigens, damaging the intestinal barrier and promoting a proinflammatory response that can lead to metabolic dysfunction and immune disorders.

Clinical Focus

- Clinical findings of gut dysbiosis.

Dysbiosis of the Colon

Clinical Findings: Local Gastrointestinal		Clinical Findings: Extraintestinal	
Gas/Flatulence Bloating Cramping Urgency Diarrhea Constipation Mucus in Stool Recurrent *C. Diff*	IBS Diverticulitis Celiac Disease Crohn's Disease Ulcerative Colitis Gallstones Gastric Cancer Colorectal Cancer	Weight Loss/Obesity (Chronic) Fatigue Poor Memory Brain Fog Autoimmunity Sleep Disorders Anxiety Depression Mood Disorders Muscle/Joint Aches Alcohol Intolerance Pruritus/Eczema Kidney Stones	Arthritis Asthma Autism Hepatic Encephalopathy Fatty Liver Fibromyalgia Hypercholesterolemia Metabolic Syndrome Diabetes Kidney Stones Multiple Sclerosis Parkinson's Disease

See: "References: Dysbiosis of the Colon – Clinical Indicators"

Dysbiosis: Representative Clinical Findings
Local Gastrointestinal

Excessive Flatulence

Bloating

Mucus in Stool

Celiac Disease

IBS

Gut Dysbiosis

Gallstones

Cramping

Constipation/Diarrhea

GERD

See: “References: Dysbiosis of the Colon – Clinical Indicators”

Dysbiosis: Representative Clinical Findings

Extraintestinal

Chronic Fatigue

Autoimmunity

Hypercholesterolemia

Weight Loss/Obesity

Anxiety/Depression

Recurrent Pruritus/Rashes

Metabolic Syndrome

Fibromyalgia

Gut Dysbiosis

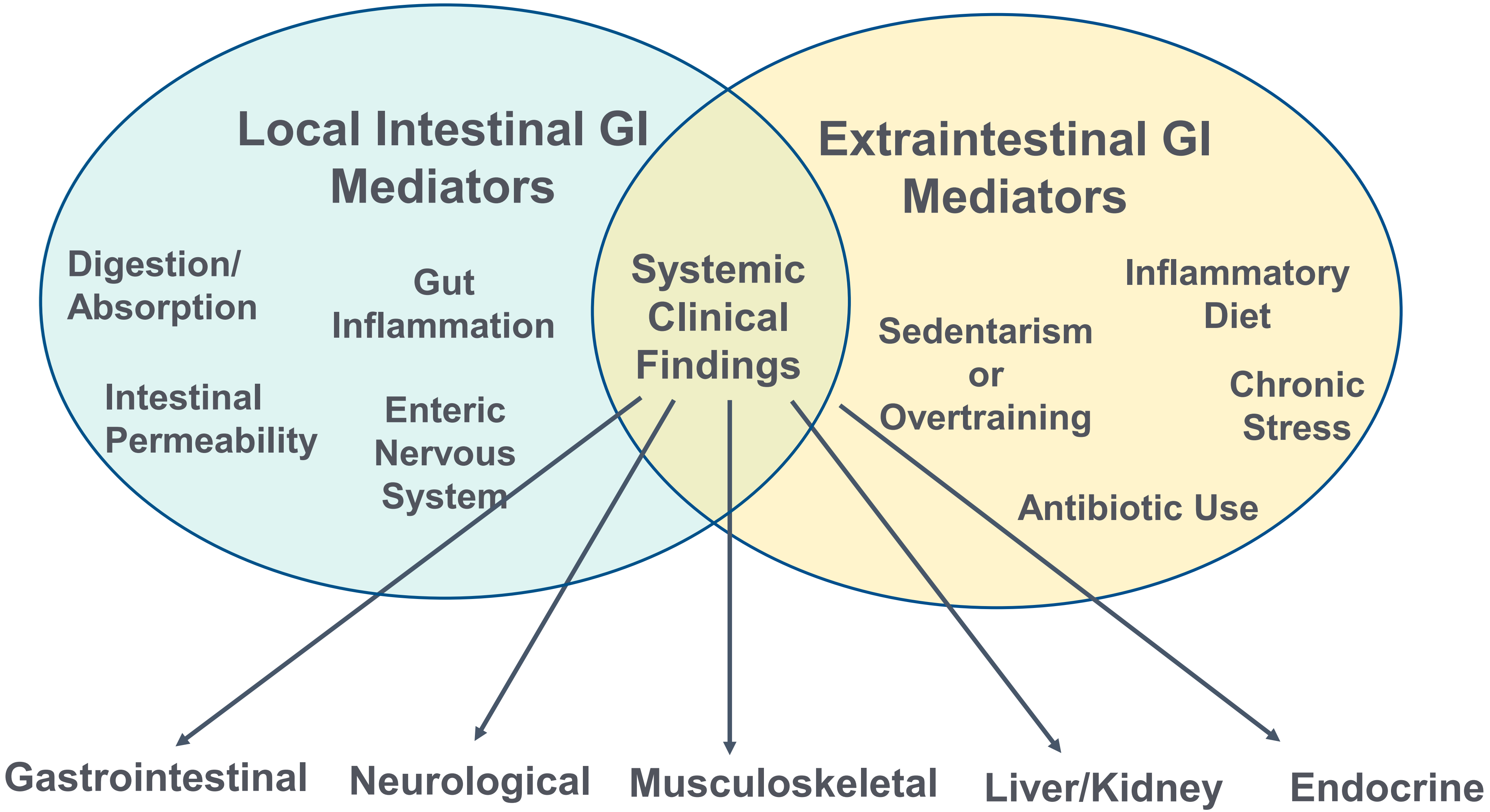
See: “References: Dysbiosis of the Colon – Clinical Indicators”

References

DYSBIOSIS OF THE COLON – CLINICAL FINDINGS

- Almario CV, Ballal ML, Chey WD, Nordstrom C, Khanna D, Spiegel BMR. Burden of gastrointestinal symptoms in the United States: results of a nationally representative survey of over 71,000 Americans. *Am J Gastroenterol*. 2018;113(11):1701-1710. doi:10.1038/s41395-018-0256-8
- Altomare A, Di Rosa C, Imperia E, Emerenziani S, Cicala M, Guarino MPL. Diarrhea predominant-irritable bowel syndrome (IBS-D): effects of different nutritional patterns on intestinal dysbiosis and symptoms. *Nutrients*. 2021;13(5):1506. doi:10.3390/nu13051506
- Halverson T, Alagiakrishnan K. Gut microbes in neurocognitive and mental health disorders. *Ann Med*. 2020;52(8):423-443. doi:10.1080/07853890.2020.1808239
- Jollet M, Nay K, Chopard A, et al. Does physical inactivity induce significant changes in human gut microbiota? New answers using the dry immersion hypoactivity model. *Nutrients*. 2021;13(11):3865. doi:10.3390/nu13113865
- Perler BK, Ungaro R, Baird G, et al. Presenting symptoms in inflammatory bowel disease: descriptive analysis of a community-based inception cohort [published correction appears in *BMC Gastroenterol*. 2020;20(1):406]. *BMC Gastroenterol*. 2019;19(1):47. doi:10.1186/s12876-019-0963-7
- Roth W, Zadeh K, Vekariya R, Ge Y, Mohamadzadeh M. Tryptophan metabolism and gut-brain homeostasis. *Int J Mol Sci*. 2021;22(6):2973. doi:10.3390/ijms22062973
- Nikhra V. The gut microbial dysbiosis, its fallouts and redressal. *J Virol Immunol*. 2019:1:1-10.

Dysbiosis: Mediators and Manifestations



See: "References: Colonic Dysbiosis – Many Root Causes”

Long Description: Dysbiosis Mediators and Manifestations

- Local intestinal G I mediators
  - Digestion
  - Absorption
  - Gut inflammation
  - Enteric nervous system
  - Intestinal permeability
- Extra-intestinal G I mediators
  - Sedentarism or overtraining
  - Antibiotic use
  - Inflammatory diet
  - Chronic stress.
- Systemic clinical findings
  - Gastrointestinal
  - Neurological
  - Musculoskeletal
  - liver, kidney
  - Endocrine

References

COLONIC DYSBIOSIS – MANY ROOT CAUSES

Antibiotic Use:

- Kumari R, Yadav Y, Misra R, et al. Emerging frontiers of antibiotics use and their impacts on the human gut microbiome. *Microbiol Res*. 2022;263:127127. doi:10.1016/j.micres.2022.127127
- Karakan T, Ozkul C, Küpeli Akkol E, Bilici S, Sobarzo-Sánchez E, Capasso R. Gut-brain-microbiota axis: antibiotics and functional gastrointestinal disorders. *Nutrients*. 2021;13(2):389. doi:10.3390/nu13020389

Chronic Stress:

- Morys J, Małecki A, Nowacka-Chmielewska M. Stress and the gut-brain axis: an inflammatory perspective. *Front Mol Neurosci*. 2024;17:1415567. doi:10.3389/fnmol.2024.1415567

Endocrine:

- Qi X, Yun C, Pang Y, Qiao J. The impact of the gut microbiota on the reproductive and metabolic endocrine system. *Gut Microbes*. 2021;13(1):1-21. doi:10.1080/19490976.2021.1894070

Gastrointestinal:

- Haneishi Y, Furuya Y, Hasegawa M, Picarelli A, Rossi M, Miyamoto J. Inflammatory bowel diseases and gut microbiota. *Int J Mol Sci*. 2023;24(4):3817. doi:10.3390/ijms24043817
- Marcari AB, Paiva AD, Simon CR, Dos Santos MESM. Leaky gut syndrome: an interplay between nutrients and dysbiosis. *Curr Nutr Rep*. 2025;14(1):25. doi:10.1007/s13668-025-00614-7
- Weiss GA, Hennet T. Mechanisms and consequences of intestinal dysbiosis. *Cell Mol Life Sci*. 2017;74(16):2959-2977. doi:10.1007/s00018-017-2509-x

Inflammatory Diet:

- Kang GG, Trevaskis NL, Murphy AJ, Febbraio MA. Diet-induced gut dysbiosis and inflammation: key drivers of obesity-driven NASH. *iScience*. 2022;26(1):105905. doi:10.1016/j.isci.2022.105905

References

COLONIC DYSBIOSIS – MANY ROOT CAUSES

Liver/Kidney:

- Tilg H, Adolph TE, Trauner M. Gut-liver axis: pathophysiological concepts and clinical implications. *Cell Metab*. 2022;34(11):1700-1718. doi:10.1016/j.cmet.2022.09.017
- Wang R, Tang R, Li B, Ma X, Schnabl B, Tilg H. Gut microbiome, liver immunology, and liver diseases. *Cell Mol Immunol*. 2021;18(1):4-17. doi:10.1038/s41423-020-00592-6
- Al Khodor S, Shatat IF. Gut microbiome and kidney disease: a bidirectional relationship. *Pediatr Nephrol.* 2017;32(6):921-931. doi:10.1007/s00467-016-3392-7

Musculoskeletal:

- Aboushaala K, Wong AYL, Barajas JN, et al. The human microbiome and its role in musculoskeletal disorders. *Genes (Basel)*. 2023;14(10):1937. doi:10.3390/genes14101937

Neurological:

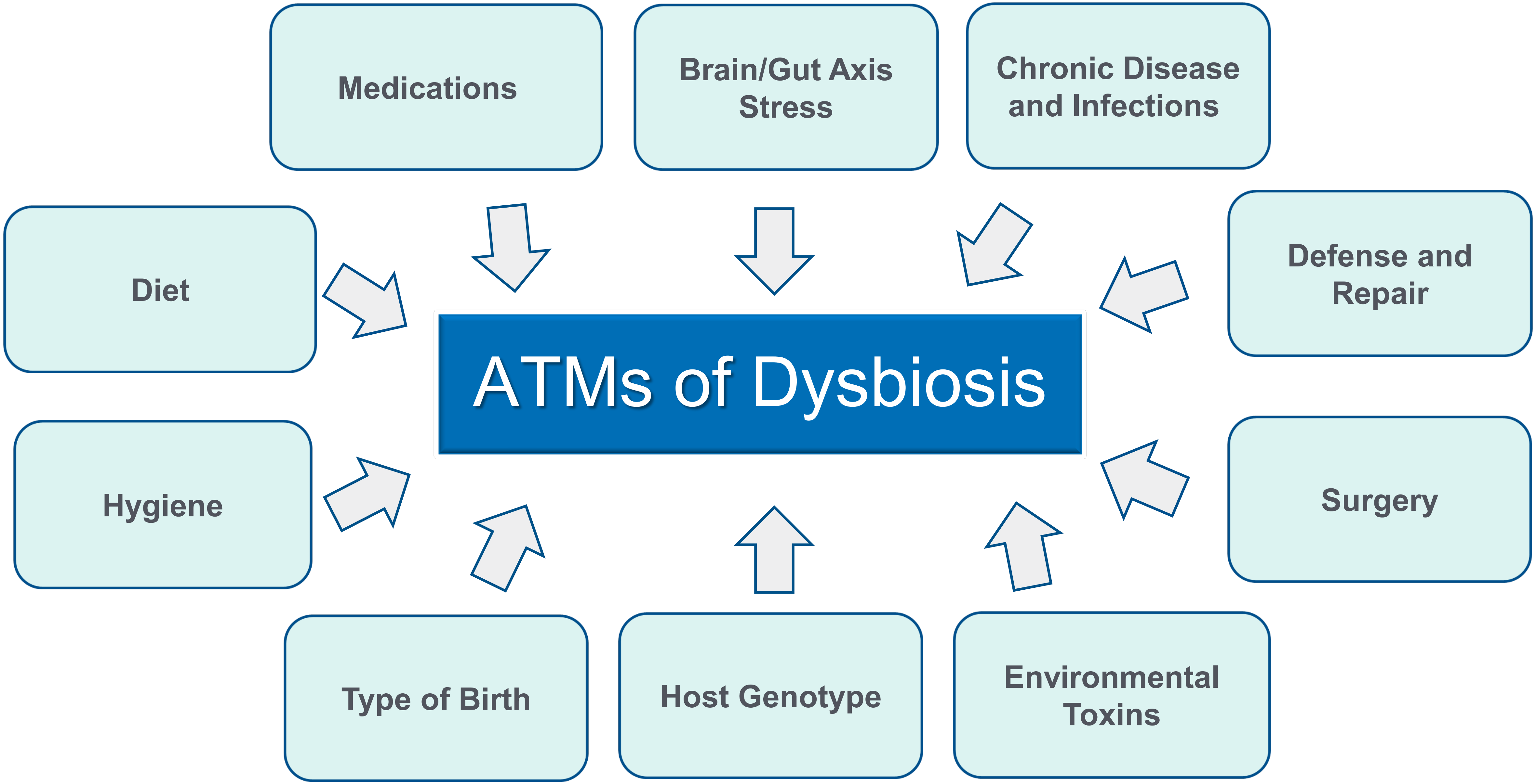
- Keshavarzian A, Sisodia SS. Gut microbiota dysbiosis and neurologic diseases: new horizon with potential diagnostic and therapeutic impact [published correction appears in *Neurotherapeutics*. 2024:e00502. doi:10.1016/j.neurot.2024.e00502]. *Neurotherapeutics*. 2024;21(6):e00478. doi:10.1016/j.neurot.2024.e00478
- Ullah H, Arbab S, Tian Y, et al. The gut microbiota-brain axis in neurological disorder. *Front Neurosci.* 2023;17:1225875. doi:10.3389/fnins.2023.1225875

Sedentarism or Overtraining:

- Patel BK, Patel KH, Lee CN, Moochhala S. Intestinal microbiota interventions to enhance athletic performance—a review. *Int J Mol Sci*. 2024;25(18):10076. doi:10.3390/ijms251810076
- Álvarez-Herms J, Odriozola A. Microbiome and physical activity. *Adv Genet*. 2024;111:409-450. doi:10.1016/bs.adgen.2024.01.002
- Przewłócka K, Folwarski M, Kaźmierczak-Siedlecka K, Skonieczna-Żydecka K, Kaczor JJ. Gut-muscle axis exists and may affect skeletal muscle adaptation to training. *Nutrients.* 2020;12(5):1451. doi:10.3390/nu12051451

Clinical Focus

- Antecedents, triggers, and mediators of gut dysbiosis.



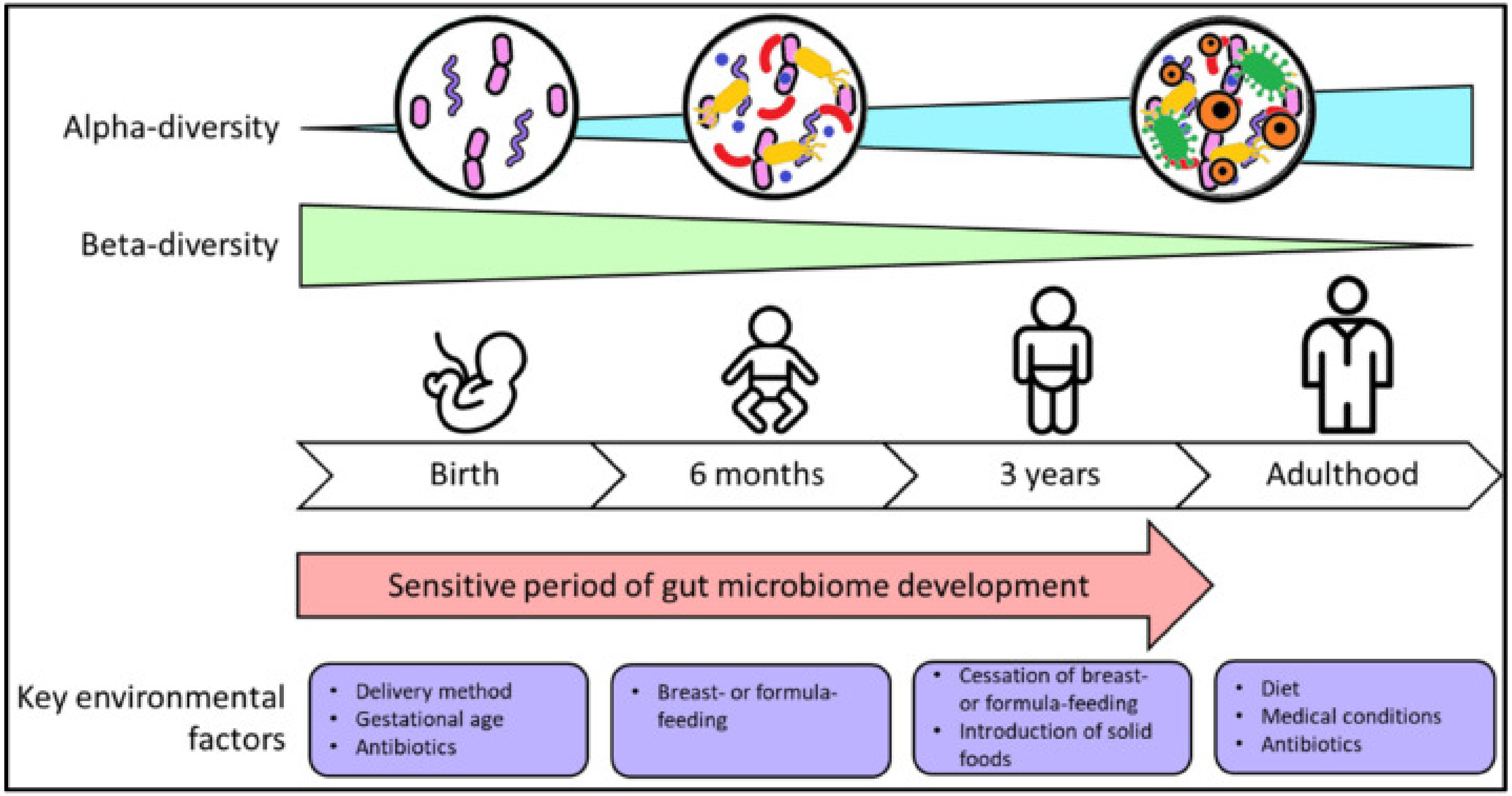
Hadrich D. Microbiome research is becoming the key to better understanding health and nutrition. *Front Genet*. 2018;9:212. doi:10.3389/fgene.2018.00212

Long Description: A T Ms of Dysbiosis

Antecedents, triggers, and mediators of dysbiosis

- Medications
- Brain-gut axis, stress
- Chronic disease and infection
- Defense and repair
- Surgery
- Environmental toxins
- Host genotype
- Type of birth
- Hygiene
- Diet

Gut Microbiome Plasticity
Antecedents, Triggers, and Mediators



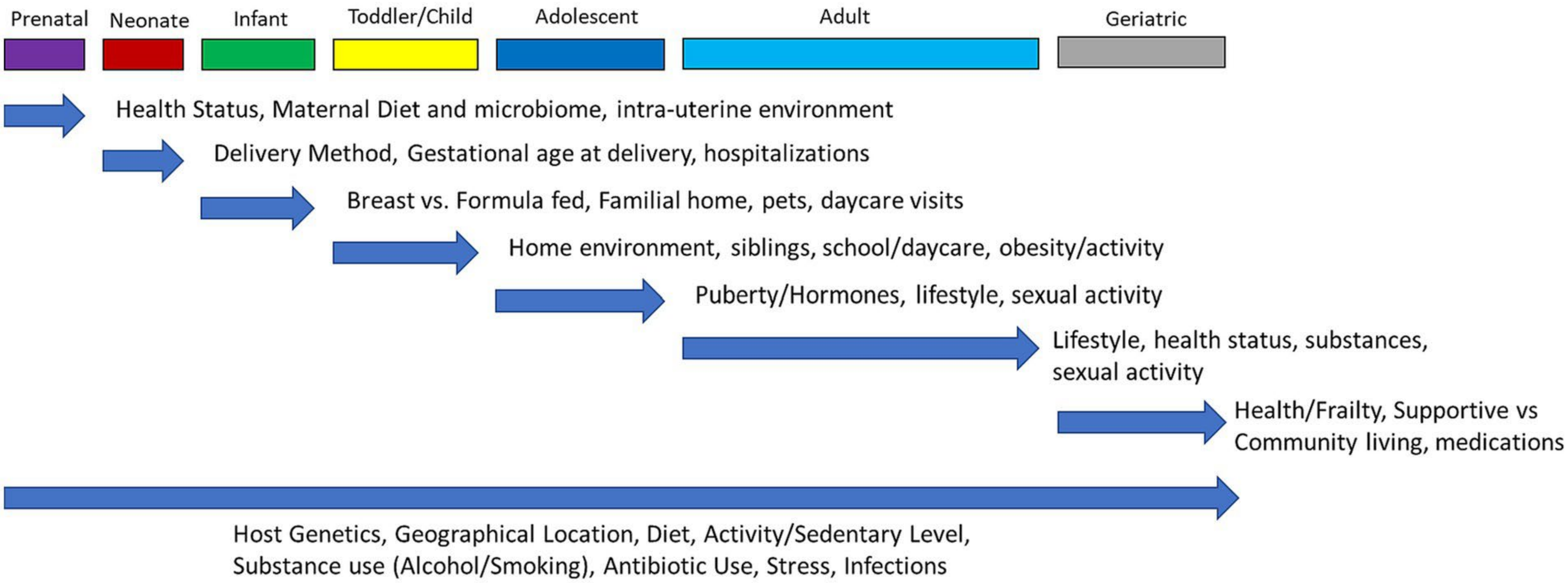
Parkin K, Christophersen CT, Verhasselt V, Cooper MN, Martino D. Risk factors for gut dysbiosis in early life. *Microorganisms.* 2021;9(10):2066. doi:10.3390/microorganisms9102066

Long Description: Gut Microbiome Plasticity

- From birth to adulthood, alpha diversity increased while beta diversity decreases. The sensitive period of gut microbiome development occurs in the first three years of life, with key environmental factors such as delivery method, gestational age, antibiotics, breast or formula feeding, cessation of breast or formula feeding, and introduction of solid foods. In adulthood, diet, medical conditions, and antibiotics influence the microbiome.

Gut Microbiome Plasticity

Antecedents, Triggers, and Mediators



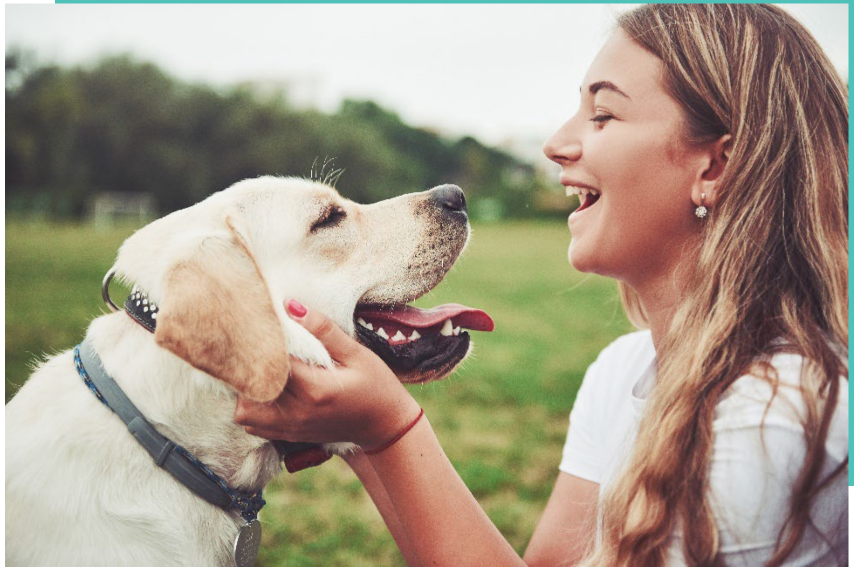
Alagiakrishnan K, Morgadinho J, Halverson T. Approach to the diagnosis and management of dysbiosis. Front Nutr. 2024 Apr 19;11:1330903. doi: 10.3389/fnut.2024.1330903. PMID: 38706561; PMCID: PMC11069313. https://creativecommons.org/licenses/by/4.0/ CC BY-4.0

Long Description: Gut Microbiome Plasticity Antecedents, Triggers, and Mediators

- The microbiome is influenced by numerous factors throughout the life course prenatally through late adulthood
- Prenatal
    - Health status
    - Maternal diet and microbiome
    - Intra-uterine environment
- Neonate
    - Delivery method
    - Gestational age at delivery
    - Hospitalizations
- Infant
    - Breast versus formula fed
    - Familial home
    - Pets
    - Daycare visits
- Toddler/Child
    - Home environment
    - Siblings
    - School/daycare
    - Obesity/activity
- Adolescent
    - Puberty/hormones
    - Lifestyle
    - Sexual activity
- Adult
    - Lifestyle
    - Health status
    - Substances
    - Sexual activity
- Geriatric
    - Health/frailty
    - Supportive versus community living
    - Medications
- Throughout the lifespan
    - Host genetics
    - Geographical location
    - Diet
    - Activity/sedentary lifestyle
    - Substance use, alcohol, smoking
    - Antibiotic use
    - Stress
    - Infections

Antecedents

Shaping the Microbiome in the Home



- Breastfeeding
- Birth delivery mode
- Hygiene hypothesis

- Pet exposure
- Washing dishes by hand
- Antibiotic use

Kim H, Sitarik AR, Woodcroft K, Johnson CC, Zoratti E. Birth mode, breastfeeding, pet exposure, and antibiotic use: associations with the gut microbiome and sensitization in children. *Curr Allergy Asthma Rep*. 2019;19(4):22. doi:10.1007/s11882-019-0851-9

Adverse Childhood Experiences

Antecedents of Dysbiosis

- Women reporting two or more adverse childhood events (ACE) had **a greater abundance of *Prevotella* and trended toward a lower abundance of *Erysipelotrichaceae* and *Phascolarctobacterium*** compared to low ACE participants.
- Abundance of several gut taxa were significantly associated with cortisol, IL-6, TNF-α, and CRP, independent of ACE.

Recent human research has associated higher levels of *Prevotella* species at mucosal surfaces with both ***localized*** and ***widespread*** disease.

- Hantsoo L, Jašarević E, Criniti S, et al. Childhood adversity impact on gut microbiota and inflammatory response to stress during pregnancy. *Brain Behav Immun*. 2019;75:240-250. doi:10.1016/j.bbi.2018.11.005
- Larsen JM. The immune response to *Prevotella* bacteria in chronic inflammatory disease. *Immunology*. 2017;151(4):363-374. doi:10.1111/imm.12760

Triggers of Dysbiosis

“Never well since…”

Antibiotic use:

- Antibiotic exposure appears to dampen anti-viral immunity and impacts the commensal bacteria

Surgery:

- Bowel resection, colonoscopy, pancreatic surgery, bariatric surgery

Infections (viral/bacterial/parasitic):

- Giardia induces alterations in the balance of the gut microbiome

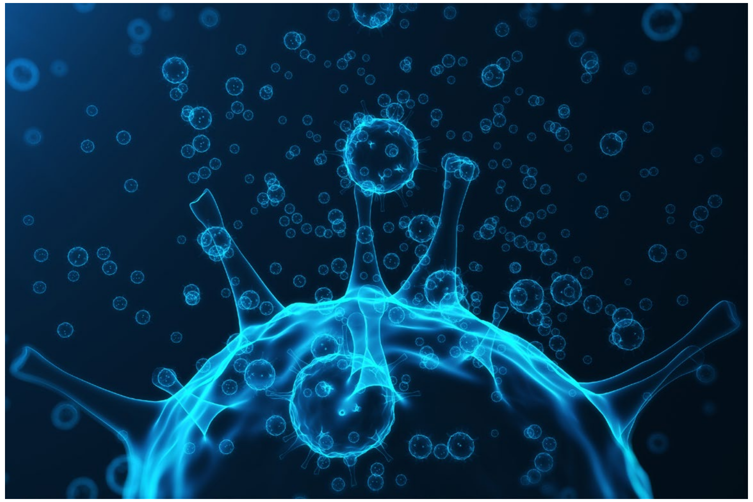
- Klimczak S, Packi K, Rudek A, et al. The influence of the Protozoan *Giardia lamblia* on the modulation of the immune system and alterations in host glucose and lipid metabolism. *Int J Mol Sci*. 2024;25(16):8627. doi:10.3390/ijms25168627
- Harper A, Vijayakumar V, Ouwehand AC, et al. Viral infections, the microbiome, and probiotics. *Front Cell Infect Microbiol*. 2021;10:596166. doi:10.3389/fcimb.2020.596166
- Tsigalou C, Paraschaki A, Bragazzi NL, et al. Alterations of gut microbiome following gastrointestinal surgical procedures and their potential complications [published correction appears in *Front Cell Infect Microbiol*. 2023;13:1281527. doi:10.3389/fcimb.2023.1281527]. *Front Cell Infect Microbiol*. 2023;13:1191126. doi:10.3389/fcimb.2023.1191126

Triggers of Gut Dysbiosis

COVID-19 Severity & Gut Microbiome

- Gut microbiome composition was significantly altered in patients with COVID-19 compared with non-COVID-19 individuals, irrespective of whether patients had received medication ($p<0.01$).

  - This correlated with: ↑ disease severity, concentrations of inflammatory cytokines, CRP, lactate dehydrogenase, aspartate aminotransferase, and GGT.
  - Gut commensals were low and remained low up to 30 days post-disease resolution.



Overall, dysbiosis following disease resolution may contribute to persistent symptoms.

Yeoh YK, Zuo T, Lui GC, et al. Gut microbiota composition reflects disease severity and dysfunctional immune responses in patients with COVID-19. *Gut*. 2021;70(4):698-706. doi:10.1136/gutjnl-2020-323020

Air Pollution

Trigger and Mediator of Gut-Lung Axis Dysfunction

Ozone exposure triggers dysbiotic changes by supporting or inhibiting microbial growth.

Dysbiotic species produce endotoxins that travel to lungs and trigger respiratory tract inflammation.

Immune cell mediators travel to lungs via mesenteric lymph, further contributing to inflammation.

There is clear cross-talk between the lung and gut microbiome; "gut-lung axis"

See: "References: Air Pollution: Trigger and Mediator of Gut-Lung Axis Dysfunction"

References

AIR POLLUTION: TRIGGER AND MEDIATOR OF GUT-LUNG AXIS DYSFUNCTION

- Fouladi F, Bailey MJ, Patterson WB, et al. Air pollution exposure is associated with the gut microbiome as revealed by shotgun metagenomic sequencing. *Environ Int*. 2020;138:105604. doi:10.1016/j.envint.2020.105604

- Zhang Z, Wang C, Lin C, et al. Association of long-term exposure to ozone with cardiovascular mortality and its metabolic mediators: evidence from a nationwide, population-based, prospective cohort study. *Lancet Reg Health West Pac*. 2024;52:101222. doi:10.1016/j.lanwpc.2024.101222

- Ni S, Yuan X, Cao Q, et al. Gut microbiota regulate migration of lymphocytes from gut to lung. *Microb Pathog*. 2023;183:106311. doi:10.1016/j.micpath.2023.106311

Toxins as Mediators of Dysbiosis
Glyphosate

- Inhibits enzyme pathway that disrupts production of amino acids

- Exposure associated with altered gut microbiota composition associated with:
  - Increases in *Bacteroides* and *Clostridium* spp.
  - Decreases in *Bifidobacterium*, *Lactobacillus*, and *Ruminococcus* spp.

See: "References: Toxins as Mediators of Dysbiosis: Glyphosate"

Toxins as Mediators of Dysbiosis
Glyphosate

Glyphosate-induced microbiota alterations associated with:

- Changes in intestinal integrity
- Lower fecal acetate levels (beneficial SCFA)
- Reproductive toxicity
- Hyperhomocysteinemia
- Neurodevelopmental disturbances
- Buildup of toxic metabolites

See: "References: Toxins as Mediators of Dysbiosis: Glyphosate"

References

TOXINS AS MEDIATORS OF DYSBIOSIS: GLYPHOSATE

- Rueda-Ruzafa L, Cruz F, Roman P, Cardona D. Gut microbiota and neurological effects of glyphosate. *Neurotoxicology*. 2019;75:1-8. doi:10.1016/j.neuro.2019.08.006
- Walsh L, Hill C, Ross RP. Impact of glyphosate (Roundup™) on the composition and functionality of the gut microbiome. *Gut Microbes.* 2023;15(2):2263935. doi:10.1080/19490976.2023.2263935
- Ueyama J, Hayashi M, Hirayama M, et al. Effects of pesticide intake on gut microbiota and metabolites in healthy adults. *Int J Environ Res Public Health*. 2022;20(1):213. doi:10.3390/ijerph20010213

Medications as Mediators of Dysbiosis

- **19 out of 52 medications** tested were found to have a significant association with changes in the gut microbiome

- Medications with the highest association:
  - **PPIs**
  - **Antibiotics**
  - **Anticholinergic inhalers**
  - **Acetaminophen**
  - **SSRIs**
  - **Opioids**
  - **Metformin**
  - **Laxatives**

See "References: Medications as Mediators of Dysbiosis”

References

MEDICATIONS AS MEDIATORS OF DYSBIOSIS

- Jackson MA, Verdi S, Maxan ME, et al. Gut microbiota associations with common diseases and prescription medications in a population-based cohort. *Nat Commun*. 2018;9(1):2655. doi:10.1038/s41467-018-05184-7
- Imhann F, Bonder MJ, Vich Vila A, et al. Proton pump inhibitors affect the gut microbiome. *Gut*. 2016;65(5):740-748. doi:10.1136/gutjnl-2015-310376
- Pant A, Maiti TK, Mahajan D, Das B. Human gut microbiota and drug metabolism. *Microb Ecol*. 2023;86(1):97-111. doi:10.1007/s00248-022-02081-x
- Petakh P, Kamyshna I, Kamyshnyi A. Effects of metformin on the gut microbiota: a systematic review. *Mol Metab*. 2023;77:101805. doi:10.1016/j.molmet.2023.101805
- Tawam D, Baladi M, Jungsuwadee P, Earl G, Han J. The positive association between proton pump inhibitors and *Clostridium difficile* infection. *Innov Pharm*. 2021;12(1):10.24926/iip.v12i1.3439. doi:10.24926/iip.v12i1.3439
- Weersma RK, Zhernakova A, Fu J. Interaction between drugs and the gut microbiome. *Gut*. 2020;69(8):1510-1519. doi:10.1136/gutjnl-2019-320204

Clinical Focus

- Representative patient example of dysbiosis on the matrix

Gut Dysbiosis & Systemic Inflammation

Dysbiosis in the colon impacts the immune system (GALT) throughout the GI tract and can trigger immune mechanisms, leading to development of a pro-inflammatory state and inflammatory diseases.

Dysbiosis disturbs metabolic function that can manifest as impaired digestive function, including:

- Abnormal breakdown of dietary components and SCFA production
- Bile acid metabolism
- Increased hydrogen sulfide production

Zhao M, Chu J, Feng S, et al. Immunological mechanisms of inflammatory diseases caused by gut microbiota dysbiosis: a review. *Biomed Pharmacother*. 2023;164:114985. doi:10.1016/j.biopha.2023.114985

Gut Dysbiosis & Systemic Inflammation

Dysbiosis contributes to GI mucosal structural changes that allow for pathogens and endotoxins to disseminate systemically, leading to inflammation outside of the digestive tract. These structural changes include:

- Intestinal epithelial cell damage
- A thinner mucus layer
- Reduced tight junction proteins

Zhao M, Chu J, Feng S, et al. Immunological mechanisms of inflammatory diseases caused by gut microbiota dysbiosis: a review. *Biomed Pharmacother*. 2023;164:114985. doi:10.1016/j.biopha.2023.114985

THE INSTITUTE FOR FUNCTIONAL MEDICINE®

Functional Medicine Matrix

Colonic Dysbiosis:

A clinical case example

The Patient's Story

Antecedents (Inherited & Acquired)

- C-section delivery

Triggers & Precipitating Events

- Parasitic infection

Mediators/Perpetuators

- Ultra-processed diet
- Environmental toxins

Clinical Imbalances and Physiological Dysfunctions

Assimilation

- IBS, gas/bloating, constipation, diarrhea

Defense & Repair

- Seasonal allergies

Mental – Emotional – Spiritual

- Anxious feelings

Structural Integrity

- Arthritis

Energy

- Fatigue, brain fog

Communication

- Hypertension

Transport

- Pre-diabetic

Mitigation & Clearance

- Metabolic Dysfunction-Associated Steatotic Liver Disease

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships
• Poor sleep	• Inactivity	• High ultra-processed foods	• Work/life balance	

Name: ______________________ Date: __________ CC: __________

Long Description: Colonic Dysbiosis

- Colonic dysbiosis clinical case
- Antecedents
  - C section delivery
- Triggers and precipitating events
  - Parasitic infection
- Mediators and perpetuators
  - Ultra-processed diet
  - Environmental toxins
- Modifiable lifestyle factors
  - Poor sleep
  - Inactivity
  - High ultra-processed foods
  - Work life balance

- Clinical Imbalances and physiological dysfunctions
  - Assimilation
    - I B S
    - Gas, bloating
    - Constipation
    - Diarrhea
  - Defense and repair
    Seasonal allergies
  - Energy
    - Fatigue
    - Brain fog
  - Mitigation and Clearance
    - Metabolic dysfunction-associated steatotic liver disease
  - Transport
    •Prediabetic
  - Communication
    Hypertension
  - Structural Integrity
    - Arthritis
  - Mental, emotional, and spiritual
    - Anxious feelings

References

COLONIC DYSBIOSIS: CLINICAL CASE EXAMPLE

- Afzaal M, Saeed F, Shah YA, et al. Human gut microbiota in health and disease: unveiling the relationship. *Front Microbiol*. 2022;13:999001. doi:10.3389/fmicb.2022.999001

Medications:

- Jackson MA, Verdi S, Maxan ME, et al. Gut microbiota associations with common diseases and prescription medications in a population-based cohort. *Nat Commun*. 2018;9(1):2655. doi:10.1038/s41467-018-05184-7

Environmental Toxins:

- Walsh L, Hill C, Ross RP. Impact of glyphosate (Roundup™) on the composition and functionality of the gut microbiome. *Gut Microbes.* 2023;15(2):2263935. doi:10.1080/19490976.2023.2263935

Parasitic Infection:

- Klimczak S, Packi K, Rudek A, et al. The influence of the Protozoan *Giardia lamblia* on the modulation of the immune system and alterations in host glucose and lipid metabolism. *Int J Mol Sci*. 2024;25(16):8627. doi:10.3390/ijms25168627

C-Section Delivery:

- Kim H, Sitarik AR, Woodcroft K, Johnson CC, Zoratti E. Birth mode, breastfeeding, pet exposure, and antibiotic use: associations with the gut microbiome and sensitization in children. *Curr Allergy Asthma Rep*. 2019;19(4):22. doi:10.1007/s11882-019-0851-9

Ultra-Processed Foods/Modifiable Lifestyle Habits:

- Hadrich D. Microbiome research is becoming the key to better understanding health and nutrition. *Front Genet*. 2018;9:212. doi:10.3389/fgene.2018.00212
- Álvarez-Herms J, Odriozola A. Microbiome and physical activity. *Adv Genet*. 2024;111:409-450. doi:10.1016/bs.adgen.2024.01.002

Energy:

- Ullah H, Arbab S, Tian Y, et al. The gut microbiota-brain axis in neurological disorder. *Front Neurosci.* 2023;17:1225875. doi:10.3389/fnins.2023.1225875

References

COLONIC DYSBIOSIS: CLINICAL CASE EXAMPLE

Pre-Diabetes:

- Vrieze A, Van Nood E, Holleman F, et al. Transfer of intestinal microbiota from lean donors increases insulin sensitivity in individuals with metabolic syndrome [published correction appears in *Gastroenterology*. 2013;144(1):250]. *Gastroenterology*. 2012;143(4):913-6.e7. doi:10.1053/j.gastro.2012.06.031

Arthritis:

- Aboushaala K, Wong AYL, Barajas JN, et al. The human microbiome and its role in musculoskeletal disorders. *Genes (Basel)*. 2023;14(10):1937. doi:10.3390/genes14101937

Gastrointestinal:

- Marcari AB, Paiva AD, Simon CR, Dos Santos MESM. Leaky gut syndrome: an interplay between nutrients and dysbiosis. *Curr Nutr Rep*. 2025;14(1):25. doi:10.1007/s13668-025-00614-7
- Weiss GA, Hennet T. Mechanisms and consequences of intestinal dysbiosis. *Cell Mol Life Sci*. 2017;74(16):2959-2977. doi:10.1007/s00018-017-2509-x

Clinical Focus

- Summary and Conclusion – Key Takeaways

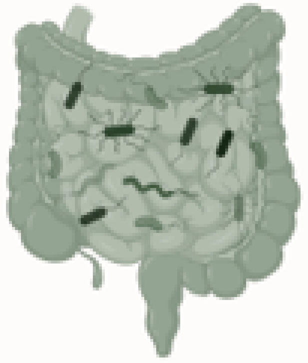
Beneficial Functions of the Gut Microbiome

STRUCTURAL
- ↑ Differentiation & proliferation of epithelial cells
- Intestinal villi & crypts development
- Villus vascularization/angiogenesis
- Tight junction regulation

METABOLISM
- SCFA production & metabolism
- Glucose metabolism
- ↑ Insulin sensitivity
- ↓ Adipose deposition
- Bone homeostasis
- Bile acid metabolism

BIOSYNTHESIS
- Hormones (e.g., CCK, 5-HT, ghrelin)
- Neurotransmitters (e.g., serotonin, GABA, catecholamines)
- Neuropeptides, enzymes
- ↑ Antioxidant production

BENEFICIAL GUT MICROBIOME FUNCTIONS

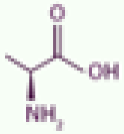


EUBIOSIS

DETOXIFICATION
- Metabolism of xenobiotics/toxins
- Drug metabolism - bioavailability/efficacy

IMMUNITY
- GALT maturation & mucosal immune regulation
- Innate & adaptive immune activation
- Immunocompetence/ tolerance
- ↓ Inflammatory mediators

NUTRITION



- Vitamin synthesis: Vit K, thiamine (B1), riboflavin (B2), biotin (B7), folate (B9), pantothenic acid (B5), cobalamin (B12)
- Hydrolysis/fermentation of complex carbohydrates
- Amino acid synthesis

PROTECTION
- ↑ Barrier integrity, mucus layer
- ↑ sIgA
- ↑ Antimicrobial peptides
- ↓ Luminal pH
- ↓ Pathogenic colonization

- Created with BioRender.com by The Institute for Functional Medicine
- See: "References: Gut Microbiome Functions"

Long Description: Beneficial Functions of the Gut Microbiome

Beneficial gut microbiome functions

- Structural
    - Increased differentiation and proliferation of epithelial cells
    - Intestinal villi and crypts development
    - Villus vascularization, angiogenesis
    - Tight junction regulation
- Immunity
    - GALT maturation and mucosal immune regulation
    - Innate and adaptive immune activation
    - Immunocompetence, tolerance
    - Decreased inflammatory mediators
- Nutrition
    - Synthesis of vitamin K, thiamine, riboflavin, biotin, folate, pantothenic acid, cobalamin
    - Hydrolysis, fermentation of complex carbohydrates
    - Amino acid synthesis
- Metabolism
    - Short chain fatty acid production and metabolism
    - Glucose metabolism
    - Increased insulin sensitivity
    - Decreased adipose deposition
    - Bone homeostasis
    - Bile acid metabolism
- Biosynthesis
    - Hormones including C C K, 5 H T, ghrelin
    - Neurotransmitters including serotonin, GABA, catecholamines
    - Neuropeptides, enzymes
    - Increased antioxidant production
- Detoxification
    - Metabolism of xenobiotics and toxins
    - Drug metabolism
- Protection
    - Increased barrier integrity, mucus layer
    - Increased secretory I G A
    - Increased antimicrobial peptides
    - Decreased luminal P H
    - Decreased pathogenic colonization

Dysbiosis of the Colon

Clinical Findings: Local Gastrointestinal

Gas/Flatulence Bloating Cramping Urgency Diarrhea Constipation Mucus in Stool Recurrent *C. Diff*	IBS Diverticulitis Celiac Disease Crohn’s Disease Ulcerative Colitis Gallstones Gastric Cancer Colorectal Cancer

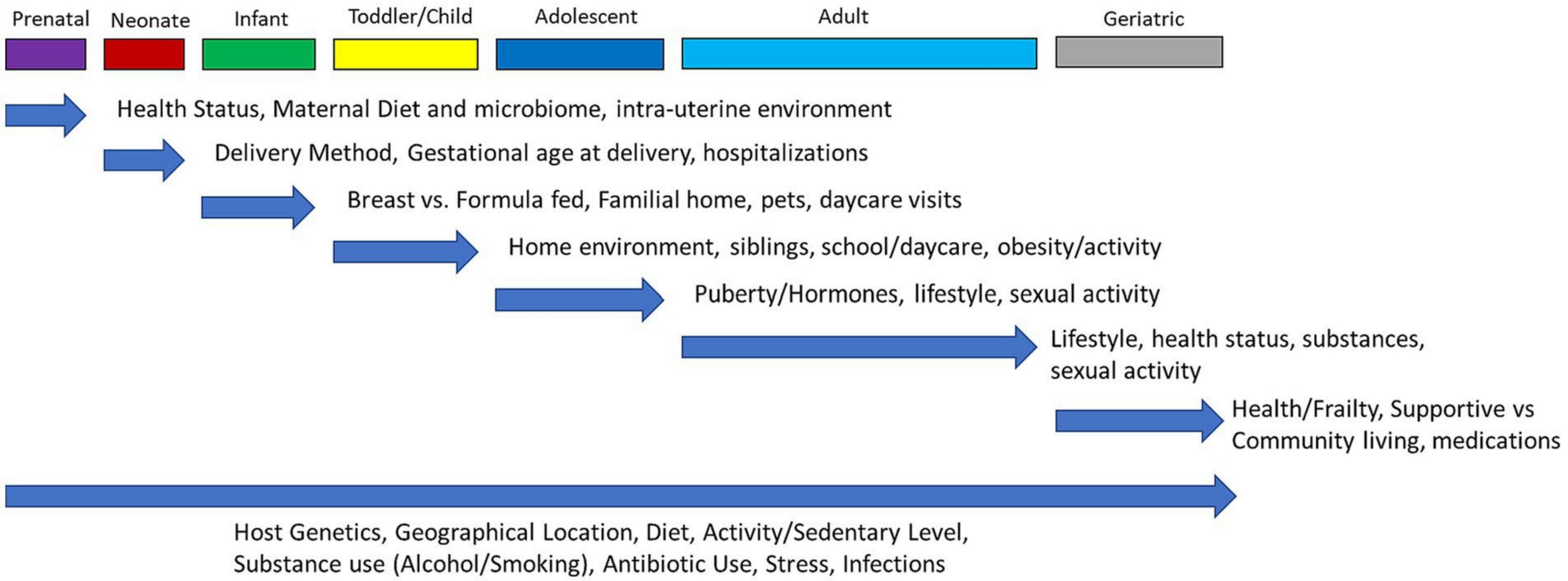
Clinical Findings: Extraintestinal

Weight Loss/Obesity (Chronic) Fatigue Poor Memory Brain Fog Autoimmunity Sleep Disorders Anxiety Depression Mood Disorders Muscle/Joint Aches Alcohol Intolerance Pruritus/Eczema Kidney Stones	Arthritis Asthma Autism Hepatic Encephalopathy Fatty Liver Fibromyalgia Hypercholesterolemia Metabolic Syndrome Diabetes Kidney Stones Multiple Sclerosis Parkinson’s Disease

See: “References “Dysbiosis of the Colon – Clinical Findings”

Gut Microbiome Plasticity

Antecedents, Triggers, and Mediators



Alagiakrishnan K, Morgadinho J, Halverson T. Approach to the diagnosis and management of dysbiosis. *Front Nutr*. 2024;11:1330903. doi:10.3389/fnut.2024.1330903

Long Description: Gut Microbiome Plasticity Antecedents, Triggers, and Mediators

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    - Familial home
    - Pets
    - Daycare visits
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    - Home environment
    - Siblings
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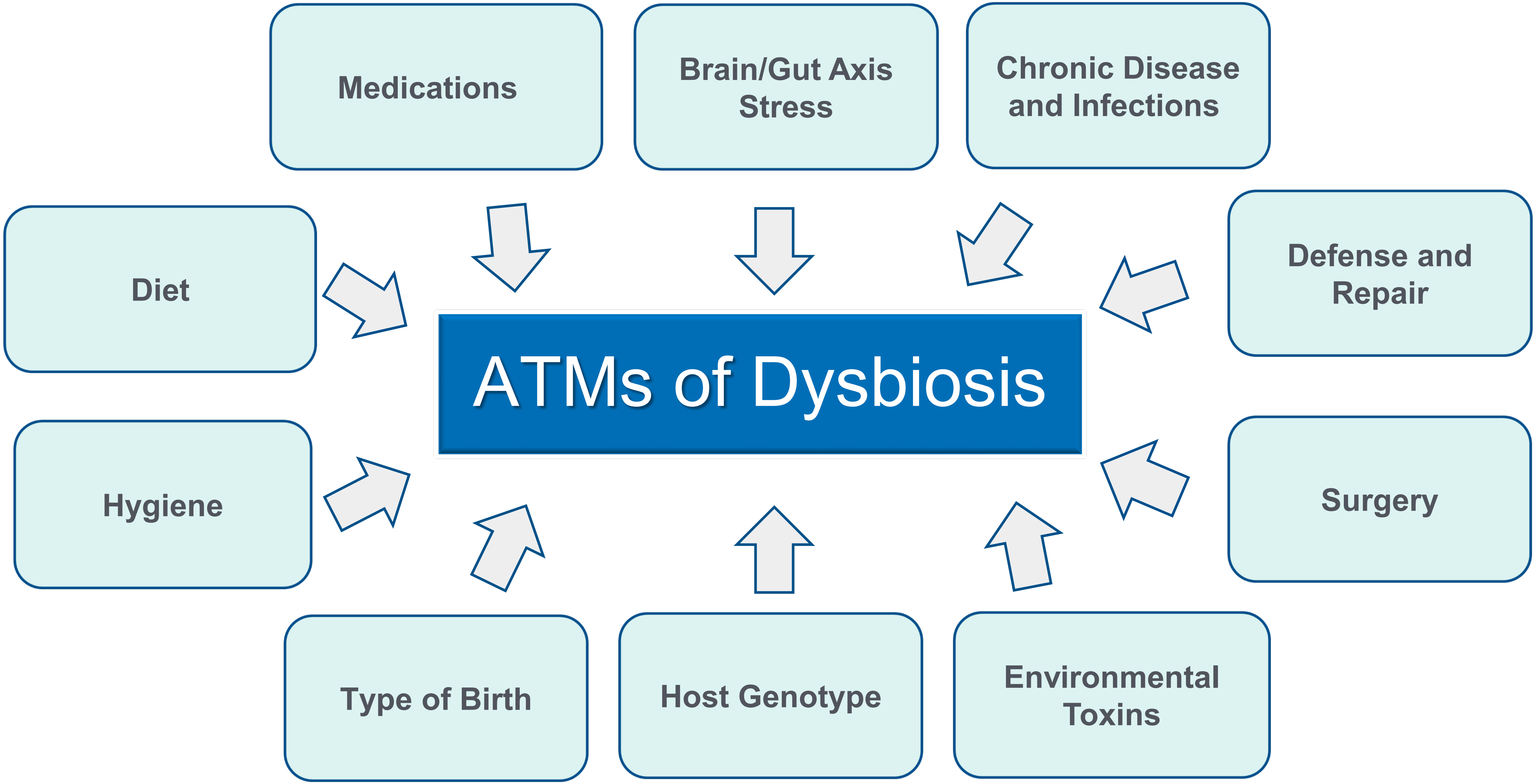
    - Obesity/activity
- Adolescent
    - Puberty/hormones
    - Lifestyle
    - Sexual activity
- Adult
    - Lifestyle
    - Health status
    - Substances
    - Sexual activity
- Geriatric
    - Health/frailty
    - Supportive versus community living
    - Medications
- Throughout the lifespan
    - Host genetics
    - Geographical location
    - Diet
    - Activity/sedentary lifestyle
    - Substance use, alcohol, smoking
    - Antibiotic use
    - Stress
    - Infections



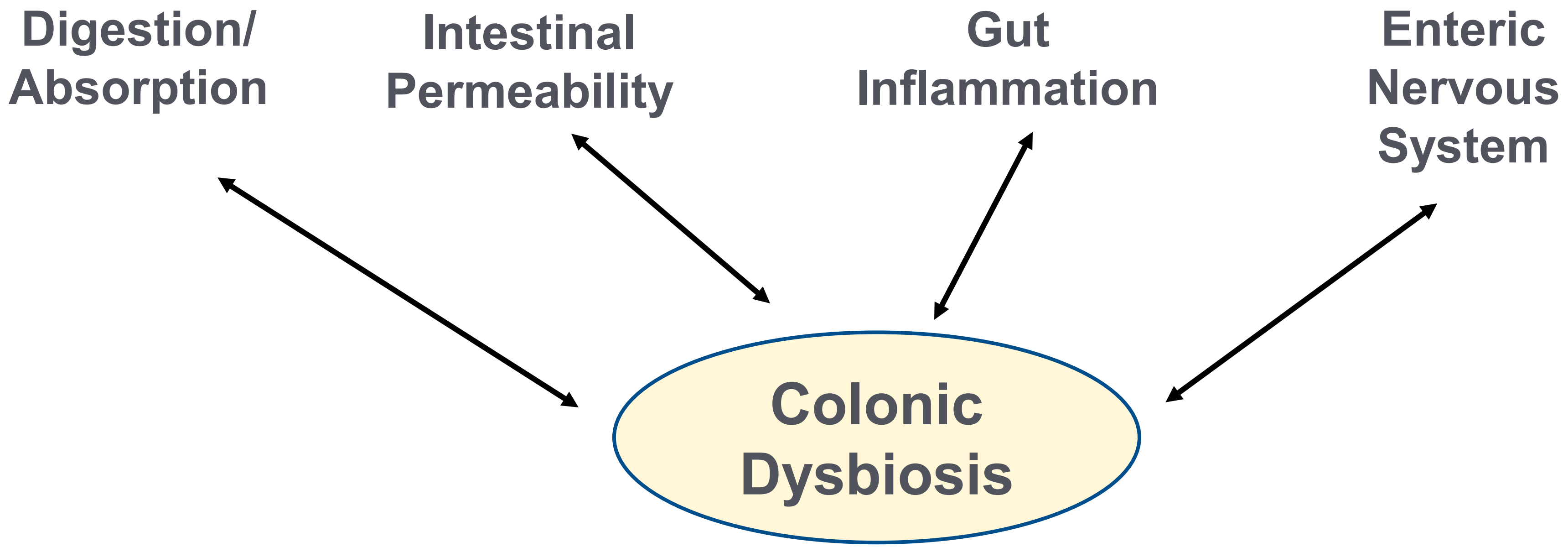
Hadrich D. Microbiome research is becoming the key to better understanding health and nutrition. *Front Genet*. 2018;9:212. doi:10.3389/fgene.2018.00212

Long Description: A T Ms of Dysbiosis

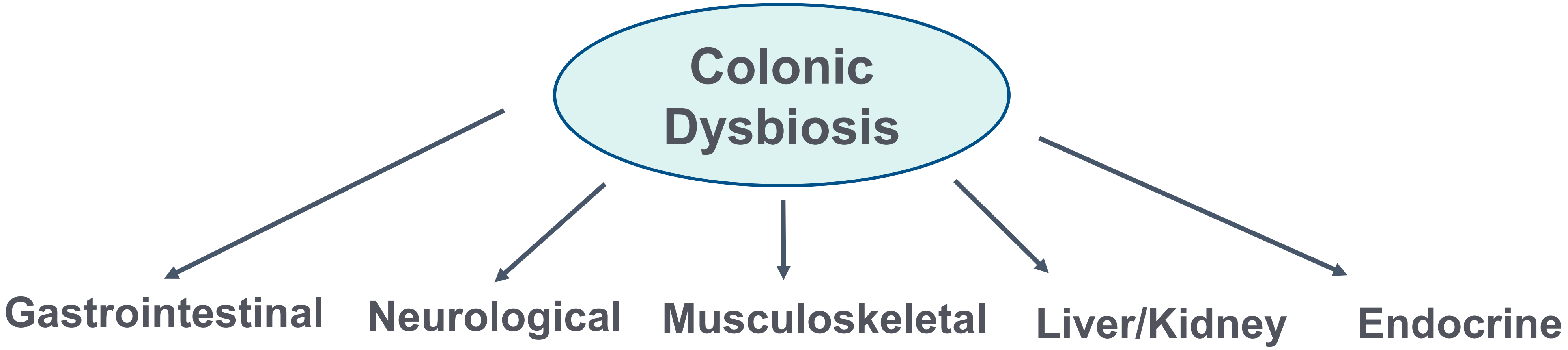
Antecedents, triggers, and mediators of dysbiosis

- Medications
- Brain-gut axis, stress
- Chronic disease and infection
- Defense and repair
- Surgery
- Environmental toxins
- Host genotype
- Type of birth
- Hygiene
- Diet

COLONIC DYSBIOSIS – LOCAL GI MEDIATORS



COLONIC DYSBIOSIS – EXTRA GI MANIFESTATIONS



See: "References: Colonic Dysbiosis – Many Root Causes"

Long Description: Colonic Dysbiosis

- Colonic dysbiosis: local G I mediators
  - Digestion, absorption
  - Intestinal permeability
  - Gut inflammation
  - Enteric nervous system
- Colonic dysbiosis: extra G I manifestations
  - Gastrointestinal
  - Neurological
  - Musculoskeletal
  - Liver, kidney
  - Endocrine

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Gut Dysbiosis:

Manifestations – Mediators – Mitigators

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE